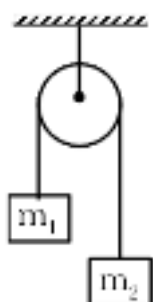


PHYSICS

1) A rocket of mass 1000 kg is to be projected vertically upwards. The gases are exhausted vertically downwards with velocity 100 m/s with respect to the rocket. What is the minimum rate of burning fuel, so as to just lift the rocket upwards against the gravitational attraction ? (Take $g = 10 \text{ m/s}^2$)

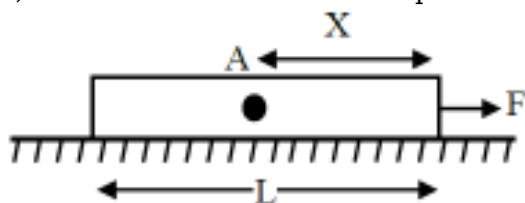
- (1) 50 kg/s
- (2) 100 kg/s
- (3) 200 kg/s
- (4) 400 kg/s

2)
A light string passing over a pulley ($m = 0$) connect two blocks of masses m_1 and m_2 . If acceleration of masses is $\frac{g}{8}$ then the ratio of $\frac{m_1}{m_2}$ is : ($m_2 > m_1$)



- (1) 8 : 1
- (2) 7 : 9
- (3) 4 : 3
- (4) 9 : 7

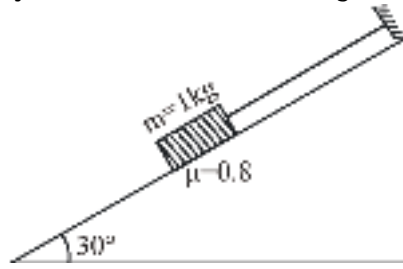
3) What is the tension in the rope at a distance x from the end where the force is applied ?



- (1) $\frac{Fx}{L-x}$
- (2) $F \frac{L}{L-x}$
- (3) $\frac{FL}{x}$

(4) $\frac{F(L-x)}{L}$

4) If the string is initially loose but almost straight then for the arrangement shown in the fig. the



tension in the string is

- (1) 6N
- (2) 6.4 N
- (3) 0.4 N
- (4) Zero

5) A block of mass m is lying on an inclined plane. The coefficient of friction between the plane and the block is μ . The force (F_1) required to move the block up the inclined plane will be :-

- (1) $mg \sin\theta + \mu mg \cos\theta$
- (2) $mg \cos\theta - \mu mg \sin\theta$
- (3) $mg \sin\theta - \mu mg \cos\theta$
- (4) $mg \cos\theta + \mu mg \sin\theta$

6) A long straight wire of radius ' a ' carries a steady current I . The current is uniformly distributed across its cross-section. The ratio of the magnetic fields at $a/2$ and $2a$ is :

- (1) $1/2$
- (2) 1
- (3) 2
- (4) $1/4$

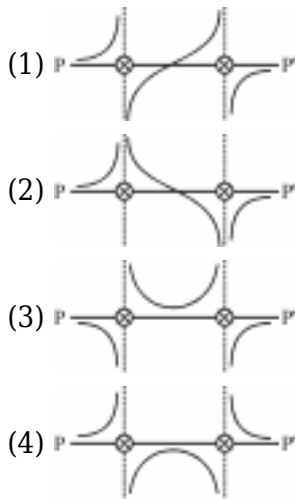
7)

The magnetic field at a distance x on the axis of a current carrying circular coil of radius R is $\frac{1}{8}$ th of that at the centre. The value of x is

- (1) $R\sqrt{3}$
- (2) $\frac{2R}{\sqrt{3}}$
- (3) $R\sqrt{2}$
- (4) $\frac{R}{\sqrt{3}}$

8) Two long parallel wire are at a distance $2d$ apart. They carry steady equal current flowing in to plane of the paper, as shown in figure. The variation of the magnetic field B along the line PP' is

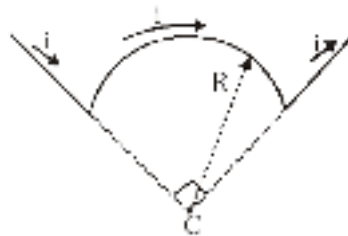
given by : (Take upwards as positive and downwards as negative).



9) A proton, deuteron and an α -particle are accelerated by same potential, enter in uniform magnetic field perpendicularly. Ratio of radii of circular path respectively :-

- (1) $1 : \sqrt{2} : \sqrt{2}$
- (2) $2 : 2 : 1$
- (3) $1 : 2 : 1$
- (4) $1 : 1 : 1$

10) The wire in the figure carries a current i and consists of a circular arc of radius R and central angle $\frac{\pi}{2}$ rad, and two straight sections whose extensions intersect the centre C of the arc. The



magnetic field that the current produces at C is :-

- (1) $|\vec{B}| = \frac{\mu_0 i}{8R}$ into the plane of the figure
- (2) $|\vec{B}| = \frac{\mu_0 i}{8R}$ out of the plane of the figure
- (3) $|\vec{B}| = \frac{\mu_0 i}{8\pi R}$ into the plane of the figure
- (4) $|\vec{B}| = \frac{\mu_0 i}{8\pi R}$ out of the plane of the figure

11) Two long straight wires are placed along x -axis and y -axis. They carry current I_1 and I_2 respectively. The equation of locus of zero magnetic induction in the magnetic field produced by them is :-

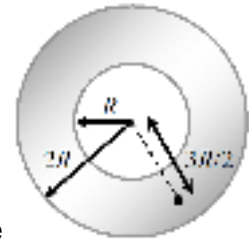
- (1) $y = x$

(2) $\left(\frac{I_2}{I_1}\right) x$

(3) $\left(\frac{I_1}{I_2}\right) x$

(4) $y = (I_1 I_2) x$

12) Figure shows the cross-sectional view of the hollow cylindrical conductor with inner radius 'R' and outer radius '2R', cylinder carrying uniformly distributed current along its axis. The magnetic



induction at point 'P' at a $\frac{3R}{2}$ distance from the axis of the cylinder will be

(1) Zero

(2) $\frac{5\mu_0 i}{72\pi R}$

(3) $\frac{7\mu_0 i}{18\pi R}$

(4) $\frac{5\mu_0 i}{36\pi R}$

13) **Statement-1** : Current carrying conductor produces only magnetic field.

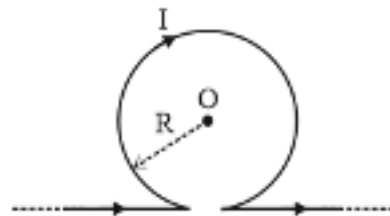
Statement-2 : Moving charge produces only magnetic field.

(1) Statement-1 is true, Statement-2 is true

(2) Statement-1 is false, Statement-2 is true

(3) Statement-1 is true, Statement-2 is false

(4) Statement-1 is false, Statement-2 is false



14) Find Magnetic field at point O in the given figure :-

(1) $\frac{\mu_0 I}{2R} \otimes$

(2) $\frac{\mu_0 I}{2R} \odot$

(3) $\frac{\mu_0 I}{2R} \left(1 - \frac{1}{\pi}\right) \otimes$

(4) $\frac{\mu_0 I}{2R} \left(1 + \frac{1}{\pi}\right) \odot$

15) A thin flexible wire of length L is connected to two adjacent fixed points carries a current I in the clockwise direction, as shown in the figure. When system is put in a uniform magnetic field of strength B going into the plane of paper, the wire takes the shape of a circle. The tension in the wire



is :

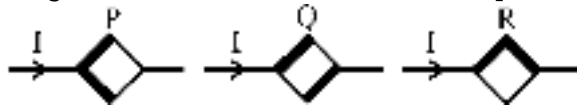
(1) IBL

(2) $\frac{IBL}{\pi}$

(3) $\frac{IBL}{2\pi}$

(4) $\frac{IBL}{4\pi}$

16) Two thick wires and two thin wires, all of the same materials and same length from a square in the three different ways P, Q and R as shown in figure with current connections as shown. The magnetic field at the centre of the square is zero in cases :-



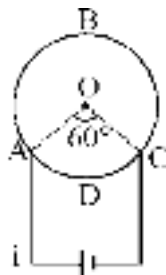
(1) P only

(2) P and Q only

(3) P and R only

(4) P, Q and R all

17) A cell is connected between the points A and C of a circular conductor ABCD of centre 'O'. $\angle AOC = 60^\circ$. If B_1 and B_2 are the magnitudes of the magnetic fields at O due to the currents in ABC and ADC respectively, the ratio $\frac{B_1}{B_2}$ is



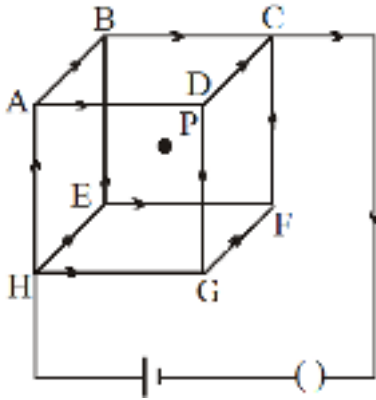
(1) 6

(2) 1

(3) 5

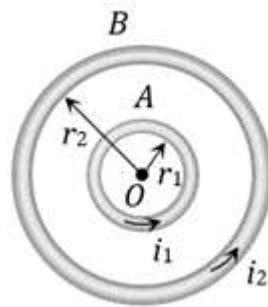
(4) 1/5

18) A steady current is set up in a cubic network composed of wires of equal resistance and length d as shown in figure. What is the magnetic field at the centre P due to the cubic network?



- (1) $\frac{\mu_0}{4\pi} \frac{2l}{d}$
- (2) $\frac{\mu_0}{4\pi} \frac{3l}{\sqrt{2}d}$
- (3) 0
- (4) $\frac{\mu_0}{4\pi} \frac{\pi l}{d}$

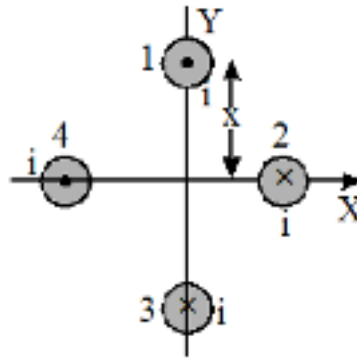
19) A and B are two concentric circular conductors of centre O and carrying current i_1 and i_2 as shown in the adjacent figure. If ratio of their radii is 1 : 2 and ratio of the flux densities at O due to A



and B is 1 : 3, then the value of i_1/i_2 is

- (1) $\frac{1}{6}$
- (2) $\frac{1}{4}$
- (3) $\frac{1}{3}$
- (4) $\frac{1}{2}$

20) What will be the resultant magnetic field at origin due to four infinite length wires if each wire



produces magnetic field 'B' at origin :-

- (1) $4B$
- (2) $\sqrt{2} B$
- (3) $2\sqrt{2} B$
- (4) Zero

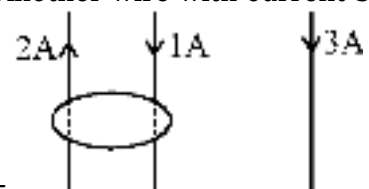
21) A charged particle moves along a circle under the action of uniform magnetic and electric fields, then this region of space may have :

- (1) $E = 0, B = 0$
- (2) $E = 0, B \neq 0$
- (3) $E \neq 0, B = 0$
- (4) $E \neq 0, B \neq 0$

22) Two straight long conductors AOB and COD are perpendicular to each other and carry currents I_1 and I_2 . The magnitude of the magnetic induction at a point P at a distance d from the point O in a direction perpendicular to the plane ABCD is :

- (1) $\frac{\mu_0}{2\pi d} (I_1 + I_2)$
- (2) $\frac{\mu_0}{4\pi d} (I_1 + I_2)$
- (3) $\frac{\mu_0}{2\pi d} \sqrt{(I_1^2 + I_2^2)}$
- (4) $\frac{\mu_0}{2\pi d} \left(\frac{I_1 I_2}{I_1 + I_2} \right)$

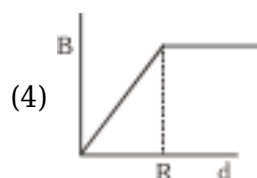
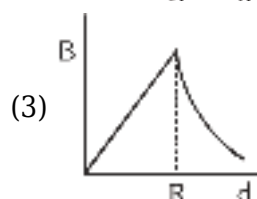
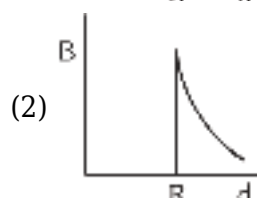
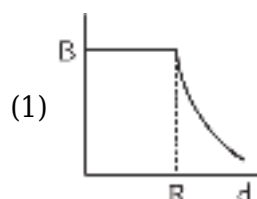
23) Two wires with currents 2 A & 1 A are enclosed in circular loop. Another wire with current 3 A is



situated outside the loop as shown. The $\oint \vec{B} \cdot d\vec{\ell}$ around the loop is :-

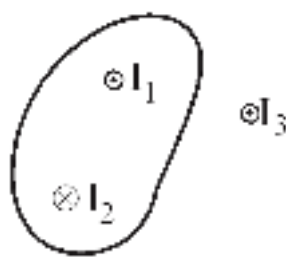
- (1) μ_0
- (2) $3\mu_0$
- (3) $6\mu_0$
- (4) $2\mu_0$

24) A solid cylindrical conductor of radius R is carrying a constant current. The plot of the magnitude of the magnetic field, B with the distance d , from the axis of the conductor, is **correctly** represented by the figure :-



25) If $\vec{B}_1$, $\vec{B}_2$ and $\vec{B}_3$ are the magnetic field due to I_1 , I_2 and I_3 , then in Ampere's circuital law

$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$, $\vec{B}$ is.



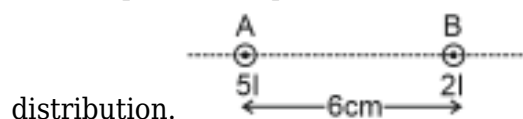
(1) $\vec{B} = \vec{B}_1 - \vec{B}_2$

(2) $\vec{B} = \vec{B}_1 - \vec{B}_2 + \vec{B}_3$

(3) $\vec{B} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3$

(4) $\vec{B} = \vec{B}_3$

26) The position of point from wire 'B', where net magnetic field is zero due to following current



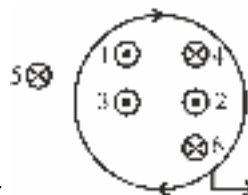
(1) $6/7$ cm

(2) $12/7$ cm

(3) $18/17$

(4) $16/7$ cm

27) The figure shown has a hypothetical amperian loop with the direction of movement marked. Six infinitely long wires having direction of current as shown and magnitude as follows are placed



perpendicular to the loop :-

$$I_2 = 2A \quad I_5 = 5A$$

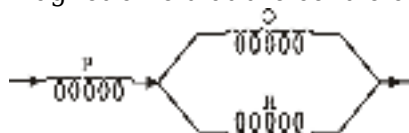
$$I_3 = 3A \quad I_6 = 4A$$

The value of $\oint \vec{B} \cdot d\vec{l}$ around the closed amperian loop will be :-

$$\text{Amperian loop: } I_1 = 1A \quad I_4 = 1A$$

- (1) μ_0 Wb/m
- (2) $-\mu_0$ Wb/m
- (3) $4\mu_0$ Wb/m
- (4) $-4\mu_0$ Wb/m

28) Three identical long solenoids P, Q and R connected to each other as shown in figure. If the magnetic field at the centre of P is 4.0 T, what would be the field at the centre of Q ?



- (1) 2.0 T
- (2) 1.0 T
- (3) 4.0 T
- (4) None of these

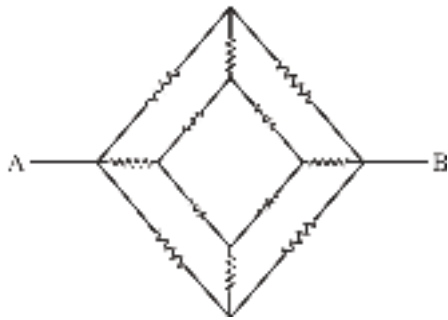
29) A solenoid of length 50 cm, having 100 turns carries a current of 2.5 A. The magnetic field at one end of the solenoid is :-

- (1) 3.14×10^{-4} T
- (2) 6.28×10^{-4} T
- (3) 1.57×10^{-4} T
- (4) 9.42×10^{-4} T

30) Mean radius of a toroid is 10 cm and number of turns are 500. If current flowing through it is 0.1 ampere then value of magnetic field (in tesla) for toroid :-

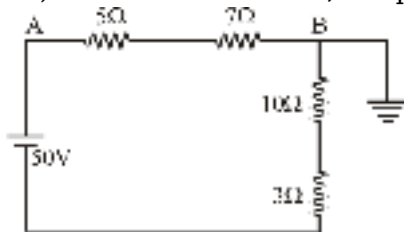
- (1) 10^{-2}
- (2) 10^{-5}
- (3) 10^{-3}
- (4) 10^{-4}

31) In given network each resistor are of 12 Ω . calculate equivalent resistance between A & B:-



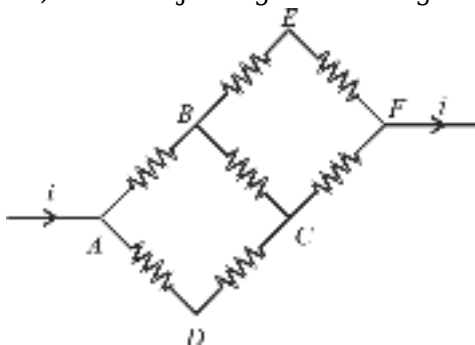
- (1) $9\ \Omega$
- (2) $\frac{12}{5}\ \Omega$
- (3) $8\ \Omega$
- (4) $\frac{11}{3}\ \Omega$

32) In the circuit shown, the point B is earthed. The potential at the point A is :-



- (1) 14 V
- (2) 24 V
- (3) 26 V
- (4) 50 V

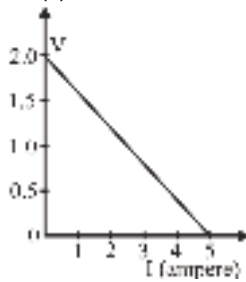
33) In the adjoining circuit diagram each resistance is of $10\ \Omega$. The current in the arm AD will be



- (1) $\frac{2i}{5}$
- (2) $\frac{3i}{5}$
- (3) $\frac{4i}{5}$
- (4) $\frac{i}{5}$

34) For a cell, the graph between the potential difference (V) across the terminals of the cell and the

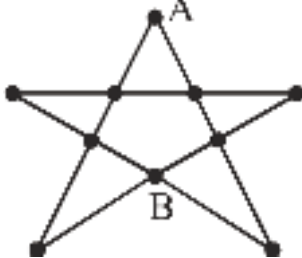
current (I) drawn from the cell is shown in the figure. the emf and the internal resistance of the cell



are :-

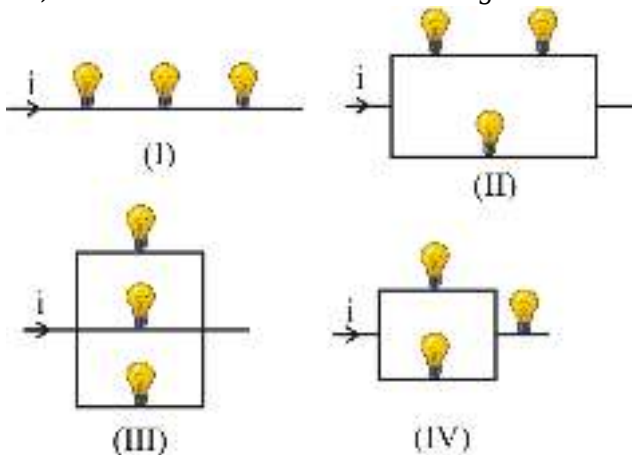
- (1) 2 V, 0.5 Ω
- (2) 2 V, 0.4 Ω
- (3) 3 V, 0.5 Ω
- (4) 3 V, 0.4 Ω

35) The resistance of all the wires between any two dots is R. The equivalent resistance between A and B in the figure is :-



- (1) $\frac{7R}{3}$
- (2) $\frac{7R}{6}$
- (3) $\frac{7R}{4}$
- (4) $\frac{7R}{5}$

36) Three identical bulbs are arranged in different combinations as shown below.



The correct order of their power consumptions is :-

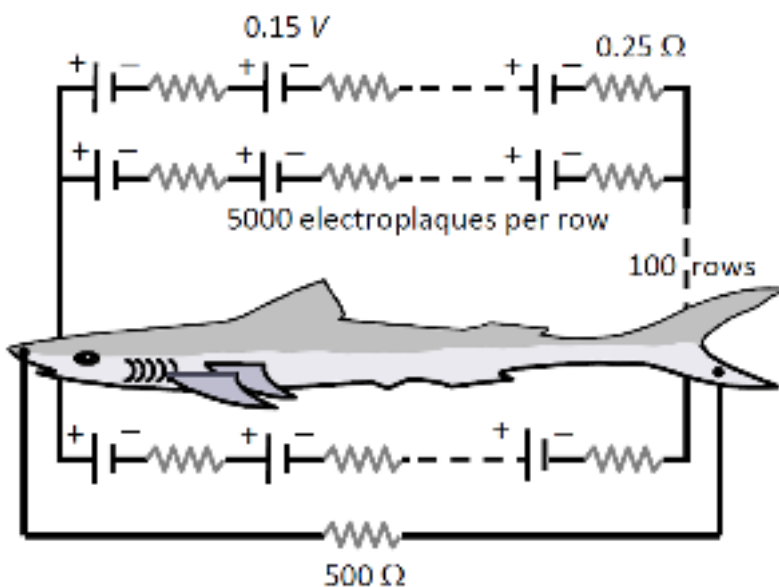
- (1) III < II < IV < I
- (2) II < III < IV < I
- (3) I < IV < III < II

(4) $I < III < II < IV$

37) To get maximum current through a resistance of 2.5Ω , one can use m rows of cells, each row having n cells. The internal resistance of each cell is 0.5Ω . What are the values of n and m if the total number of cells is 45 ?

- (1) $m = 3, n = 15$
- (2) $m = 5, n = 9$
- (3) $m = 9, n = 5$
- (4) $m = 15, n = 3$

38) Eels are able to generate current with biological cells called electroplaques. The electroplaques in an eel are arranged in 100 rows, each row stretching horizontally along the body of the fish containing 5000 electroplaques. The arrangement is suggestively shown below. Each electroplaques has an emf of 0.15 V and internal resistance of 0.25Ω .



The water surrounding the eel completes a circuit between the head and its tail. If the water surrounding it has a resistance of 500Ω , the current an eel can produce in water is about

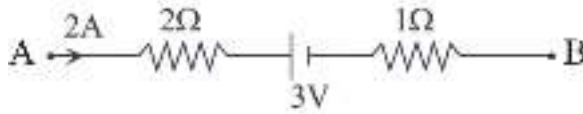
- (1) 1.5 A
- (2) 3.0 A
- (3) 15 A
- (4) 30 A

39) n identical cells are joined in series with its two cells A and B in the loop with reversed polarities. EMF of each cell is E and internal resistance r . Potential difference across cell A or B is (here $n > 4$)

- (1) $\frac{2E}{n}$
- (2) $2E \left(1 - \frac{1}{n}\right)$
- (3) $\frac{4E}{n}$

(4) $2E \left(1 - \frac{2}{n}\right)$

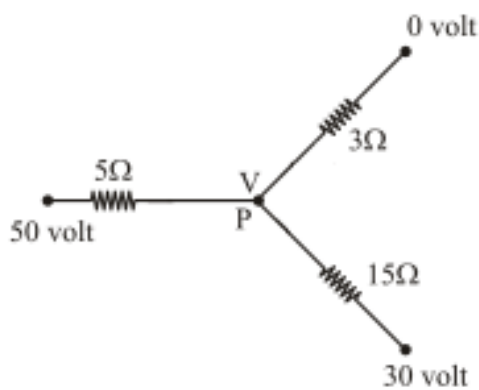
40) Figure represents a part of the closed circuit. The potential difference between points A and



B ($V_A - V_B$) is

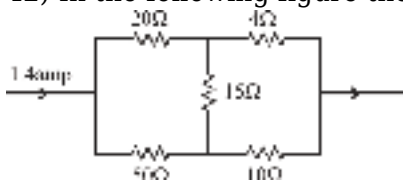
- (1) + 9 V
- (2) - 9 V
- (3) + 3 V
- (4) + 6 V

41) For the given circuit as shown in figure. The value of V at point P :-



- (1) 10 V
- (2) 20 V
- (3) 5 V
- (4) 7.5 V

42) In the following figure the current through 4 ohm resistor is :-



- (1) 1.4 amp
- (2) 0.4 amp
- (3) 1.0 amp
- (4) 0.7 amp

43) The resistivity of a wire :-

- (1) Increases with the length of the wire
- (2) Decreases with the area of cross-section
- (3) Decreases with the length and increases with the cross-section of wire

(4) None of the above statement is correct

44) If a copper wire is stretched to make it 0.1% longer. The percentage change in its resistance is :-

- (1) 0.2% increase
- (2) 0.2% decrease
- (3) 0.1% increase
- (4) 0.1% decrease

45)

The total current supplied to the circuit by the battery is :-



- (1) 1 A
- (2) 2 A
- (3) 4 A
- (4) 6 A

CHEMISTRY

1) **Assertion :-** In friedel craft's reaction of 1-chloropropane with benzene gives isopropyl benzene as product.

Reason :- The primary carbocation rearranges to form secondary carbocation.

- (1) Both assertion & reason are true & the reason is a correct explanation of the assertion
- (2) Both assertion & reason are true but reason is NOT a correct explanation of the assertion.
- (3) Assertion is true but the reason is false
- (4) Both assertion & reason are false

2) Match the pair & select appropriate option :-

Column-I Electrophillic substitution reaction		Column-II Reagent	
A.	Halogenation	P.	$\text{CH}_3\text{Cl}, \text{AlCl}_3$
B.	Friedel Craft alkylation	Q.	$\text{Cl}_2, \text{AlCl}_3$
C.	Friedel Crafts acylation	R.	$\text{H}_2\text{S}_2\text{O}_7$
D.	Sulphonation	S.	$\text{Ph}-\text{C}(=\text{O})\text{Cl}, \text{AlCl}_3$

- (1) A-Q, B-P, C-R, D-S

- (2) A-P, B-Q, C-R, D-S
 (3) A-P, B-Q, C-S, D-R
 (4) A-Q, B-P, C-S, D-R

3) **Statement I :-** In case of aryl halides, halogens are deactivating due to strong -I effect.

Statement II :- Halogens are meta directing groups.

In the light of above statements, select the correct option.

- (1) Both statement-I & II are correct
 (2) Both statement-I & II are incorrect
 (3) Statement-I is correct but II is incorrect
 (4) Statement-II is correct but I is incorrect

4) Match the following reactants in column-I with the corresponding reaction products in column-II :-

Column-I		Column-II	
(i)	Benzene + $\text{Cl}_2 \xrightarrow{\text{AlCl}_3}$	(a)	Benzoic acid
(ii)	Benzene + $\text{CH}_3\text{Cl} \xrightarrow{\text{AlCl}_3}$	(b)	Methyl phenyl ketone
(iii)	Benzene + $\text{CH}_3\text{COCl} \xrightarrow{\text{AlCl}_3}$	(c)	Toluene
(iv)	Toluene $\xrightarrow{\text{KMnO}_4/\text{NaOH}}$	(d)	Chlorobenzene

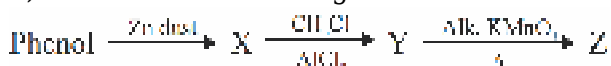
- (1) (i)-d, (ii)-c, (iii)-b, (iv)-a
 (2) (i)-d, (ii)-b, (iii)-c, (iv)-a
 (3) (i)-d, (ii)-c, (iii)-a, (iv)-b
 (4) (i)-c, (ii)-d, (iii)-a, (iv)-b

5) **Assertion :** In case of phenol, bromination takes place even in absence of lewis acid whereas bromination of benzene takes place in presence of lewis acid like FeBr_3 .

Reason : -OH group attached to benzene ring is highly deactivating.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) Assertion is True but the Reason is False.
 (4) Both Assertion & Reason are False.

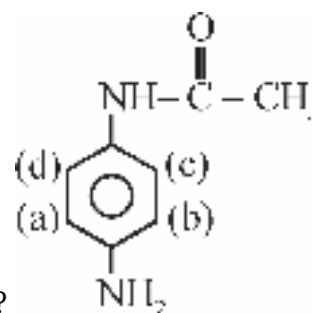
6) Consider the following reaction :



Product 'Z' is :-

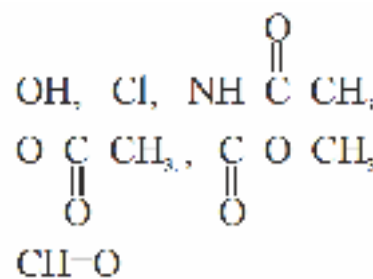
- (1) Benzaldehyde
 (2) Benzoic acid
 (3) Toluene

(4) Benzene



7) Identify position where ESR is most favourable?

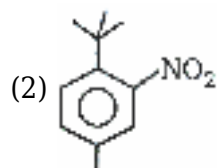
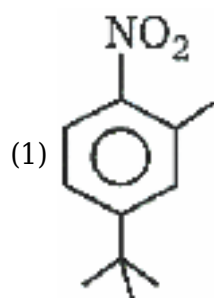
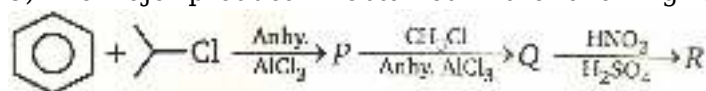
- (1) (a) and (b)
- (2) (a) only
- (3) (c) and (d)
- (4) (a), (b), (c) and (d)

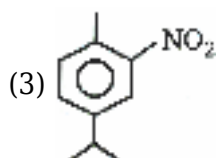


8) How many of following are ortho-para directing?

- (1) 4
- (2) 5
- (3) 6
- (4) 3

9) The major product 'R' obtained in the following reaction is :





(4) None of these

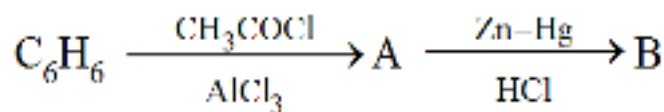
10) The correct order of reactivity towards electrophilic substitution is

(1) benzoic acid > chlorobenzene > benzene > phenol

(2) benzoic acid > phenol > benzene > chlorobenzene

(3) phenol > benzene > Chlorobenzene > benzoic acid

(4) phenol > chlorobenzene > benzene > benzoic acid



11) The end product in the below sequence is:

(1) Toluene

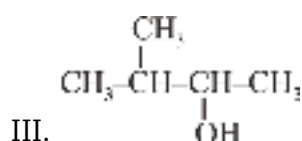
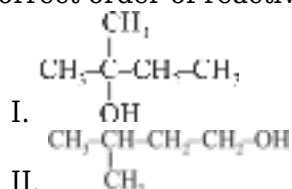
(2) Ethyl benzene

(3) Both the above

(4) None

12)

Correct order of reactivity for acidic dehydration of following alcohols

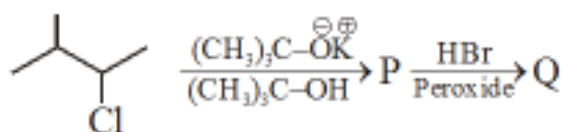


(1) I > III > II

(2) III > II > I

(3) I > II > III

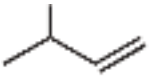
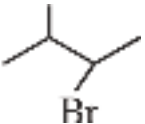
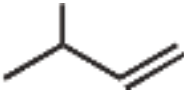
(4) II > III > I

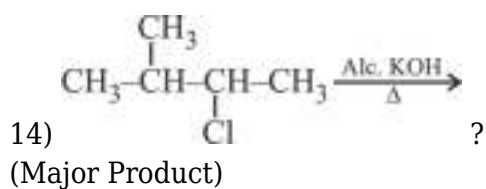


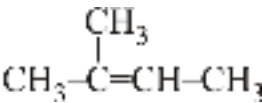
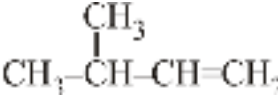
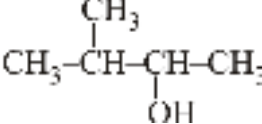
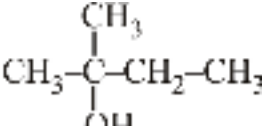
13)

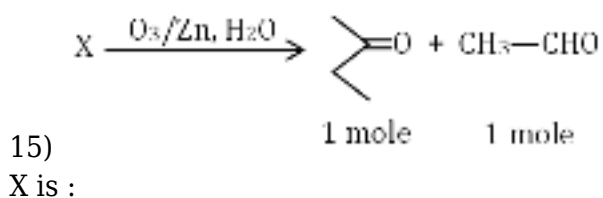
Product P and Q are :-

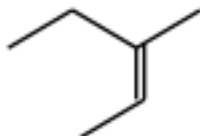
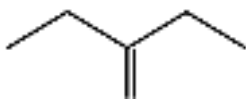
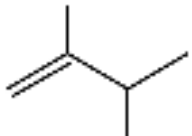


- (2)  and 
- (3)  and 
- (4)  and 

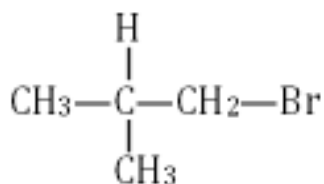


- (1) 
- (2) 
- (3) 
- (4) 



- (1) 
- (2) 
- (3) 
- (4) 

16) Arrange the following alkyl halides in decreasing order of the rate of elimination reaction with alcoholic KOH.



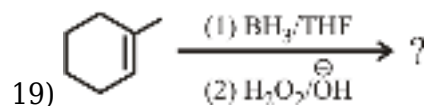
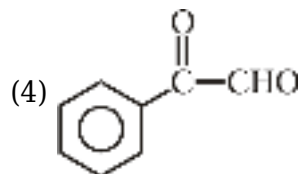
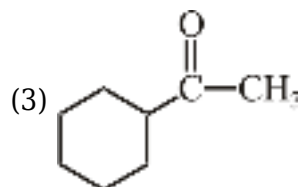
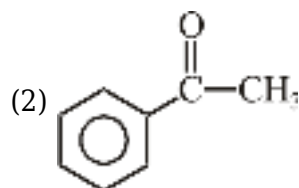
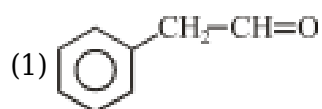
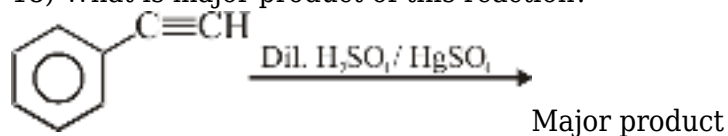
- (A)
 (B) $\text{CH}_3 - \text{CH}_2 - \text{Br}$
 (C) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{Br}$

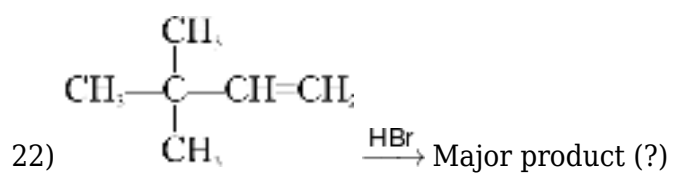
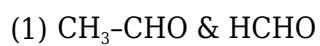
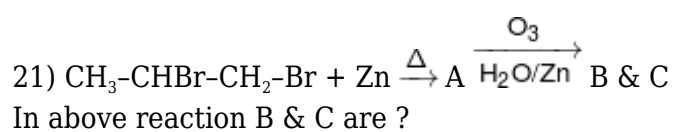
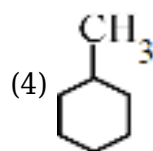
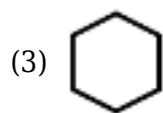
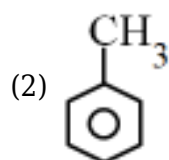
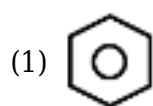
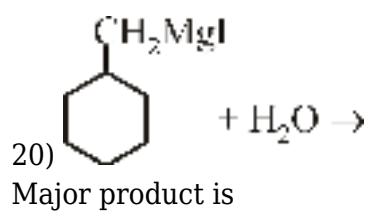
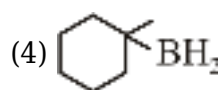
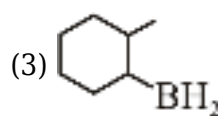
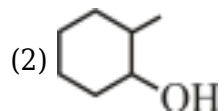
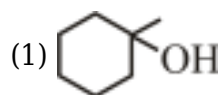
- (1) $A > B > C$
 (2) $C > B > A$
 (3) $B > C > A$
 (4) $A > C > B$

17) Statement-I : Hydrogenation of 2-Butyne in presence of Pd/BaSO_4 gives trans-2-Butene
 Statement-II : Trans-2-butene is more stable than cis-2-butene

- (1) Both Statement I and Statement II are true.
 (2) Both Statement I and Statement II are false.
 (3) Statement I is true but Statement II is false.
 (4) Statement I is false but Statement II is true.

18) What is major product of this reaction?





- (1) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH}_2 - \text{CH}_3 \\ | \quad | \\ \text{CH}_3 \quad \text{Br} \end{array}$
- (2) $\begin{array}{c} \text{CH}_3 \quad \text{Br} \\ | \quad | \\ \text{CH}_3 - \text{C} - \text{CH} - \text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$
- (3) $\begin{array}{c} \text{Br} \\ | \\ \text{CH}_3 - \text{C} - \text{CH} - \text{CH}_3 \\ | \quad | \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$
- (4) $\begin{array}{c} \text{Br} \\ | \\ \text{CH}_3 - \text{C} - \text{CH}_2 - \text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$

23)

The time for half life period of a certain reaction $A \rightarrow \text{Products}$ is 1 hour, when the initial concentration of the reactant 'A', is 2.0 mol L^{-1} , how much time does it take for its concentration to come from 0.50 to 0.25 mol L^{-1} if it is a zero order reaction ?

- (1) 0.25h
(2) 1h
(3) 4h
(4) 0.5h

24) A large increase in the rate of a reaction for a rise in temperature is due to....

- (1) the decrease in the number of collisions.
(2) the increase in the number of activated molecules.
(3) the shortening of the mean free path.
(4) the increasing of the activation energy.

25) Which reaction characteristics are changing by the addition of a catalyst to a reaction at constant temperature.....

- (i) Activation energy
(ii) Equilibrium constant
(iii) Reaction entropy
(iv) Reaction enthalpy

- (1) (i) Only
(2) (iii) only
(3) (i) and (ii) only
(4) All of these

26) For the reaction $3A + 2B \rightarrow C$, which statement is correct :-

- (1) Rate of formation of C is three times of rate of disappearance of A
- (2) Rate of disappearance of B is $\frac{3}{2}$ times of rate of disappearance of A
- (3) Rate of disappearance of A is $\frac{3}{2}$ times of rate of disappearance of B
- (4) Rate of disappearance of B is half of the rate of formation of C

27) A study of chemical kinetics of the reaction $A + B \rightarrow \text{products}$, gave the following data at 25°C :-

Exp No.	[A]	[B]	Rate
(i)	1.0	0.15	4.2×10^{-6}
(ii)	2.0	0.15	8.4×10^{-6}
(iii)	1.0	0.20	5.6×10^{-6}

Find out rate law

- (1) $r = K [A]$
- (2) $r = K[B]$
- (3) $r = K [A] [B]$
- (4) $r = K[A][B]^2$

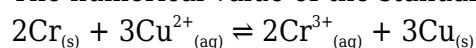
28) In the presence of acid, the initial conc. of cane sugar was reduced from 0.2 to 0.1 M in 5 hours and to 0.05 M in 10 hours from start. The reaction must be of :-

- (1) Zero order
- (2) First order
- (3) Third order
- (4) Fractional order

29) Consider the given data

	Half cell reaction	Standard reduction potential E^0 (volts)
(a)	$\text{Cr}_2\text{O}_7^{2+} + 14\text{H}^+ + 6\text{e}^- \rightleftharpoons 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	1.33 V
(b)	$\text{Cr}^{3+} + 3\text{e}^- \rightleftharpoons \text{Cr}$	- 0.74 V
(c)	$\text{Cu}^+ + \text{e}^- \rightleftharpoons \text{Cu}$	0.52 V
(d)	$\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu}$	0.34 V

The numerical value of the standard cell potential for the reaction



- (1) - 1.08 V
- (2) - 0.40 V
- (3) + 1.08 V

(4) + 2.50 V

30) The pressure of H_2 required to make the potential of H_2 -electrode zero in pure water at 298 K is :-

- (1) 10^{-14} atm
- (2) 10^{-12} atm
- (3) 10^{-10} atm
- (4) 10^{-4} atm

31) **Assertion :** Increase in the concentration of copper ion in Daniel cell increases the emf of the cell.

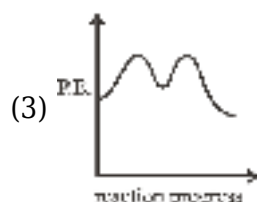
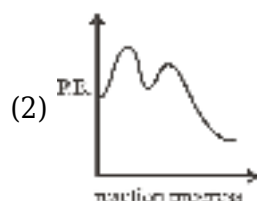
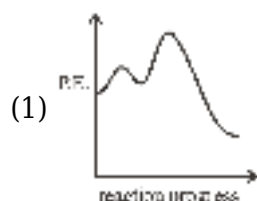
Reason : According to the nernst equation emf of Daniel cell = $E_{\text{cell}}^0 + \frac{0.059}{2} \log \left[\frac{Cu^{+2}}{Zn^{+2}} \right]$

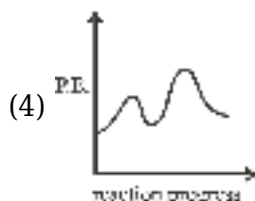
- (1) Assertion and Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Assertion and Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion and Reason are False.

32) The rate of a reaction is approximately doubled for every 10°C rise in temperature. If the temperature is raised by 50°C , the reaction rate increases by about

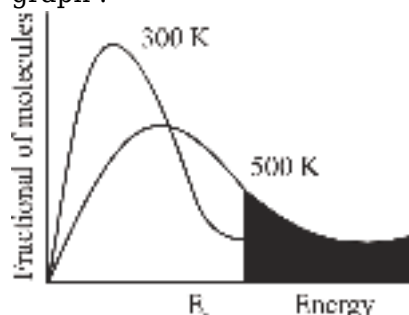
- (1) 10 times
- (2) 16 times
- (3) 32 times
- (4) 64 times

33) Select the correct diagram for an exothermic reaction that proceeds through two steps, with the second step is rate determining :-





34) The distribution of molecular kinetic energy at two temperature is as shown in the following graph :



Which of the following conclusions are correct ?

- (1) the number of molecules with energy E_a or greater is proportional to the shaded area for each temperature
- (2) The number of molecules with energy E_a or less is proportional to the shaded area for each temperature
- (3) The number of molecules with energy E_a is the mean of all temperature
- (4) The graph is not according to the Maxwell-Boltzmann energy distribution law

35) Select the correct statement regarding catalyst.

- (a) Small amount of catalyst can catalyse a large amount of reactants.
- (b) A catalyst does not alter Gibbs energy change (ΔG) of a reaction.
- (c) A catalyst can catalyses both spontaneous and non-spontaneous reaction
- (d) A catalyst catalyses the forward reaction to the more extent than a backward reaction

- (1) a and c only
- (2) a and b only
- (3) c and d only
- (4) b and d only

36) For a reaction, the rate constant is expressed as $k = Ae^{-40000/T}$. The energy of the activation is :-

- (1) 40000 cal
- (2) 88000 cal
- (3) 80000 cal
- (4) 8000 cal

37) The resistnace of 1N solution of CH_3COOH is 250Ω , when measured in a cell of cell constant is 1.15 cm^{-1} . The equivalent conductance will be

- (1) $4.6 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$
- (2) $9.2 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$

- (3) $18.4 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$
 (4) $0.023 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$

38) Which of the following solutions has the highest equivalent conductance ?

- (1) 0.01 M NaCl
 (2) 0.05 M NaCl
 (3) 0.005 M NaCl
 (4) 0.02 M NaCl

39) Molar conductivity of a solution of an electrolyte AB_3 is $150 \text{ Scm}^2 \text{ mol}^{-1}$. If it ionises as $\text{AB}_3 \rightarrow \text{A}^{3+} + 3\text{B}^-$, its equivalent conductivity will be :-

- (1) 150 (in $\text{Scm}^2 \text{ eq}^{-1}$)
 (2) 75 (in $\text{Scm}^2 \text{ eq}^{-1}$)
 (3) 50 (in $\text{Scm}^2 \text{ eq}^{-1}$)
 (4) 80 (in $\text{Scm}^2 \text{ eq}^{-1}$)

40) Calculate molar conductance at infinite dilution for acetic acid, given :-

$$\lambda_m^\infty (\text{HCl}) = 425 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$\lambda_m^\infty (\text{NaCl}) = 188 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$\lambda_m^\infty (\text{CH}_3\text{COONa}) = 96 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

- (1) $524 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
 (2) $284 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
 (3) $333 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
 (4) $123 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$

41) A given cell reaction is spontaneous when :

- (1) $E^\circ_{\text{Red.}}$ is negative
 (2) $E_{\text{Red.}}$ is positive
 (3) E°_{cell} is positive
 (4) E_{cell} is positive

42) Saturated solution of KNO_3 is used to make 'Salt-bridge' because.

- (1) Velocity of K^+ is greater than that of NO_3^-
 (2) Velocity of NO_3^- is greater than that of K^+
 (3) Velocities of both K^+ and NO_3^- are nearly the same
 (4) KNO_3 is highly soluble in water

43) If $E^\circ_{\text{Fe}^{+2}/\text{Fe}}$ is x_1 , $E^\circ_{\text{Fe}^{+3}/\text{Fe}}$ is x_2 then $E^\circ_{\text{Fe}^{+3}/\text{Fe}^{+2}}$ will be.

- (1) $3x_2 - 2x_1$

- (2) $x_2 - x_1$
- (3) $x_2 + x_1$
- (4) $2x_1 + 3x_2$

44) The reduction potential of hydrogen electrode when placed in a buffer solution is found to be -0.591 V. The pH of the buffer is :

- (1) 10
- (2) 4
- (3) 7
- (4) 12

45) Standard EMF of a galvanic cell involving $2e^-$ transfer is found to be 0.591 V at 25°C . Calculate equilibrium constant for the cell reaction

- (1) 20^{10}
- (2) 10^{20}
- (3) 10^{200}
- (4) 200^{10}

BIOLOGY

1) Which of the following is an autoimmune disorder in which antibodies are produced against acetylcholine receptors ?

- (1) Chronic hepatitis
- (2) Addison's disease
- (3) Rheumatoid arthritis
- (4) Myasthenia gravis

2) *Plasmodium* enters the human body as :-

- (1) female *Anopheles* mosquito
- (2) sporozoite
- (3) trophozoite
- (4) haemozoin

3) LSD is very powerful hallucinogenic drug obtained from :-

- (1) *Papaver somniferum*
- (2) *Claviceps purpurea*
- (3) *Cannabis sativa*
- (4) *Erythroxylum coca*

4) Which of the following can be the probable mechanism for inducing cancer :-

- (1) Overactivity of tumor suppressor gene
- (2) Overactivity of telomerase
- (3) Overactivity of telomerase inhibitor
- (4) Down regulation of proto-oncogene

5) Mark the correctly matched options :-

(A) Leaves of Cannabis sativa



(B) Opium poppy



(C) Flowing branch of Datura

- (1) A and B
- (2) B and C
- (3) A, B and C
- (4) Only C

6) Which of the following is not obtained from plant cannabis sativa?

- (1) Hashish
- (2) Charas
- (3) Cocaine
- (4) Marijuana

7) The side effects of use of anabolic steroids in male don't include :-

- (1) Increased aggressiveness
- (2) Mood swings
- (3) Liver dysfunction
- (4) Excessive hair growth on face & body

8) How many of the following statements is correct?

- (a) benign tumours show the property of metastasis
- (b) heroin accelerates body functions
- (c) malignant tumours may exhibit metastasis
- (d) patients who have undergone surgery are given cannabinoids to relieve pain.

- (1) Three

- (2) Four
- (3) Two
- (4) One

9) Read the following statements and select the correct answer.

- I. Psychoactive drugs have the ability to alter the activity of the nervous system.
- II. Addiction is a psychological attachment to certain effects such as euphoria and temporary feeling of well-being associated with drugs and alcohol.
- III. When the drugs are taken for a purpose other than medicinal use or in amounts/frequency that impairs one's physical, physiological or psychological functions, it constitutes drug abuse.
- IV. Smoking increases carbon monoxide content in blood and reduces the concentration of oxygen. This causes oxygen deficiency in the body.

- (1) I and II
- (2) III and IV
- (3) I, II and IV
- (4) I, II, III and IV

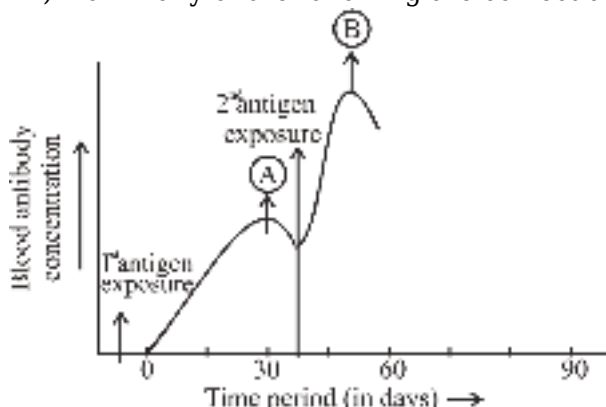
10) Opioids are the drugs which bind to specific opioid receptors present in our CNS and GIT. Heroin, commonly called smack is chemically, which is white, odourless, bitter crystalline compound, obtained by.....of morphine :-

- (1) Diacetylmorphine, methylation
- (2) Diacetylmorphine, acetylation
- (3) Benzodiazepines, amination
- (4) Amphetamines, acetylation

11) Identify the **correct** statement regarding communicable diseases ?

- (1) Mary Mallon associated with tetanus infection.
- (2) AIDS never get transmitted through artificial insemination.
- (3) Primary host of plasmodium is mosquito.
- (4) Filariasis causes chronic inflammation of blood vessels.

12) How many of the following are correct about given diagram



- (a) (A) is anamnestic response
- (b) Memory cells are responsible for (A)

(c) Memory cells are responsible for (B)

(d) IgM is responsible for (A)

- (1) One
- (2) Two
- (3) Three
- (4) Four

13) Microenvironment for development and maturation of T-lymphocytes is provided by

- (1) Only thymus
- (2) Only bone marrow
- (3) Both thymus and bone marrow
- (4) Thymus and spleen

14) How many of the followings are secondary lymphoid organs.

Spleen, Thymus, MALT, Liver, peyer's patches, Tonsils, Lymphatic vessels, Appendix

- (1) Two
- (2) Three
- (3) Four
- (4) Five

15) **Assertion (A) :-** Colostrum from mother is essential for new born babies.

Reason (R) :- Colostrum contains antibodies.

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is not correct.
- (3) (A) is not correct but (R) is correct.
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

16) **Statement-I :** Condoms are disposable, can be self inserted and thereby gives privacy to the user.

Statement-II : Progesterone alone or in combination with estrogen can be used by females as injection or implant under the skin.

- (1) Statement I and II both are correct
- (2) Statement I and II both are incorrect
- (3) Only Statement I is correct
- (4) Only Statement II is correct

17) **Statement-I :** Infertile couples are unable to produce children inspite of unprotected sexual cohabitation.

Statement-II : Medical termination of pregnancy is legalised in India.

In the light of above statements, select the correct answer from the options given below :-

- (1) Both statement-I and statement-II are correct

- (2) Statement-I is correct and statement-II is incorrect
- (3) Statement-I is incorrect and statement-II is correct
- (4) Both statement-I and statement-II are incorrect

18) **Statement-I** : Our laws permit legal adoption and it is as yet, one of the best methods for couples looking for parenthood.

Statement-II : The reasons for infertility could be many-physical, congenital, diseases, drugs, immunological or even psychological.

Choose the correct answer from the given option :-

- (1) Both **Statement I** and **Statement II** are incorrect.
- (2) Both **Statement I** and **Statement II** are correct.
- (3) **Statement I** is correct but **Statement II** is not correct
- (4) **Statement I** is incorrect but **Statement II** is correct

19) Which of the following is a technique of direct introduction of gametes into the oviduct :

- (1) MTS
- (2) ET
- (3) GIFT
- (4) POST

20) Though all persons are vulnerable to STIs but their incidences are reported to be very high among persons of age group

- (1) 25-35 years
- (2) 5-10 years
- (3) 15-24 years
- (4) 40-50 years

21) Which of the following is the correct reason for delivering sex education at schools ?

- (1) Discouraging myths and misconceptions
- (2) Encouraging rise in population
- (3) Solve the issue of global warming
- (4) Promote myths

22) How many statements are correct for IUDs :

- (i) They are self inserted
- (ii) They are inserted by Doctors or expert nurses.
- (iii) Non medicated IUDs are made up of copper.
- (iv) These devices are implanted in uterus through vagina
- (v) It is most widely accepted in India.
- (vi) IUDs are ideal contraceptives for the female who want to delay pregnancy.
- (vii) IUDs increase phagocytosis of sperms in the uterus.
- (viii) CuT, LNG-20, Vaults, cervical caps are some popular IUDs.

- (1) Four
- (2) Five
- (3) Six
- (4) Seven

23) What is true for an ideal contraceptive ?

- I. It should be user-friendly.
- II. It should be easily available.
- III. It should be ineffective and reversible with least side effects.
- IV. It should be effective and reversible with least side effects.
- V. It should interfere with the sexual act of the user.

- (1) All
- (2) I, II, III
- (3) I, II, IV
- (4) I, II, IV, V

24) How many statements are true about "Saheli"?

- (a) New oral contraceptive for the female
- (b) It is once a week pill
- (c) It contains a non-steroidal preparation
- (d) Very high side effects and low contraceptive value
- (e) It was developed by scientist at Central Drug Research Institute in Lucknow.

- (1) Three
- (2) Four
- (3) Two
- (4) Five

25) Which method of contraception has high failure rate ?

- (1) Barrier method
- (2) IUD
- (3) Sterilization
- (4) Natural method

26) How many of the following methods are included in invitro fertilization technique?
GIFT, AI, ZIFT, IUT, ICSI

- (1) 2
- (2) 1
- (3) 3
- (4) 4

27) How many of the following are copper releasing IUD's?
LNG-20, CuT, Cu7, Progestasert, Multiload-375

- (1) 2
- (2) 1
- (3) 3
- (4) 4

28) Amniocentesis is a technique

- (1) by which the essential amino acids in the body can be estimated
- (2) by which any chromosomal abnormalities in the foetus can be detected
- (3) in which the sex of the foetus can be reversed
- (4) that can be used for correcting genetic and non-genetic disorders of the foetus

29) Medical Termination of Pregnancy (MTP) is considered safe up to have many week of pregnancy?

- (1) Eight weeks
- (2) Twelve weeks
- (3) Eighteen weeks
- (4) Six weeks

30) Contraceptive oral pills help in birth control by :-

- (1) Killing the sperms in uterus
- (2) Forming barriers between sperms and ova
- (3) Preventing ovulation
- (4) Killing the ova

31) Statement-I :- Hymen is always torn during coitus.

Statement-II :- Clitoris is a tiny finger like structure present at the junction of two labia majora.

- (1) Both statement-I and statement-II are correct.
- (2) Both statement-I and statement-II are incorrect.
- (3) Statement-I is incorrect and statement-II is correct.
- (4) Statement-I is correct and statement-II is incorrect.

32) Birth canal is formed by :

- (1) Cervical canal + Uterus
- (2) Cervical canal + Vagina
- (3) Cervical canal + Isthmus
- (4) Cervical canal + Fallopian tube

33)

Which of the following is mismatched ?

- (1) Fimbriae → Finger like projection of infundibulum
- (2) Cervical canal → The cavity of cervix
- (3) Hymen → True sign of virginity
- (4) Vagina → Fibromuscular and nonglandular

34)

Identify the hormones that are secreted in large amount prior to ovulation :

A	LH	B	FSH
C	Estrogen	D	Progesterone

- (1) A only
- (2) A & B only
- (3) A, B & C only
- (4) A, B, C & D

35)

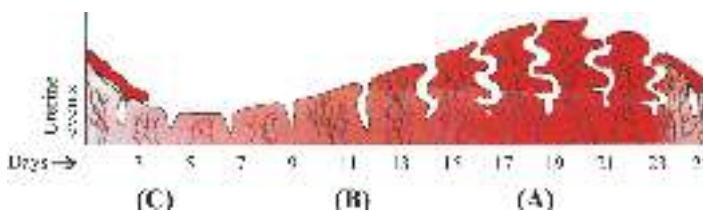
Read the following term's explanations and find out correct match :-

- (a) Gestation → Embryonic development
- (b) Amenorrhoea → Absence of menstrual cycle
- (c) Hysterectomy → Surgical removal of ovary
- (d) Oophorectomy → Surgical removal of uterus

- (1) a, b, c
- (2) a, b, c, d
- (3) a, b
- (4) a, b, d

36) Which of the following is not correct ?

- (1) Gynecomastia - Development of mammary glands in male
- (2) Ectopic pregnancy - Uterus
- (3) Oophorectomy - Removal of ovaries
- (4) Castration - Crushing of testes



37)

Choose correct option :-

	A	B	C
(1)	Menstruation	Luteal	Secretory

(2)	Luteal	Secretory	Menstruation
(3)	Secretory phase	Proliferative phase	Menstruation
(4)	Mestruation	Luteal	Secretory

(1) 1

(2) 2

(3) 3

(4) 4

38) Which structures form the extratesticular pathways for sperms?

(1) Seminiferous tubules, Epididymis, Vas deferens

(2) Seminiferous tubules, Vasa efferentia, Epididymis

(3) Epididymis, Vas deferens, Ejaculatory duct

(4) Vasa efferentia, Epididymis, Vas deferens

39) How many testicular lobules are present in each testis ?

(1) 250

(2) 350

(3) 750

(4) 400

40) For normal fertility, how many percentage of total sperms must have normal shape and size ?

(1) 50 %

(2) 24 %

(3) 40 %

(4) 60 %

41) Read the following statement and select the correct fillings ?

TheI.... is the male external genitalia. It is made up ofII.... that helps inIII.... of the penis to facilitateIV.... The enlarged end of penis called theV.... is covered by a loose fold of skin calledVI.... .

(1) I-Penis, II-corpus striatum, III-insemination, IV-Erection, V-foreskin.

(2) I-Hemipenis, II-corpus adiposum, III-Erection, IV-insemination, V-Glans penis, VI-prepace.

(3) I-Penis, II-Erectile tissue, III-Erection, IV-insemination, V-Glans penis, VI-fore skin.

(4) I-Clitoris, II-Special tissue, III-Erection, IV-insemination. V - Glans penis, VI-fore skin.

42) Correct match the column-I to column-II :

	Column-I		Column-II
--	-----------------	--	------------------

(i)	Syngamy	(a)	Intermixing of cytoplasm of sperm and ovum
(ii)	Plasmogamy	(b)	Union of male and female gametes
(iii)	Karyogamy	(c)	Intermixing of chromosomes of gametes
(iv)	Amphimixis	(d)	Fusion of pronuclei of sperm and ovum

(i) (ii) (iii) (iv)

- (1) (b) (a) (d) (c)
 (2) (b) (a) (c) (d)
 (3) (a) (b) (c) (d)
 (4) (a) (b) (d) (c)

(1) 1

(2) 2

(3) 3

(4) 4

43) **Assertion** : All copulations do not lead to fertilisation and pregnancy.

Reason : Fertilisation can only occur if the ovum and sperms are transported simultaneously to isthmus region.

- (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
 (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
 (3) Assertion is true but Reason is false.
 (4) Both Assertion and Reason are false.

44) Which among the following has 23 chromosomes?

- (1) Spermatogonia
 (2) Zygote
 (3) Secondary oocyte
 (4) Oogonia

45) Which fact about the mammary glands in humans is false?

- (1) A non-functional mammary gland is the characteristic of all male mammals.
 (2) Mammary glands are paired structures.
 (3) It is a glandular tissue containing fixed amount of fat.
 (4) Glandular tissue of each breast is divided into 15 to 20 mammary lobes containing clusters of cells called alveoli.

46) Which of the following is a blood cholesterol lowering agent, produced by *Monascus purpureus* ?

- (1) Cyclosporin-A
- (2) Statins
- (3) Streptokinase
- (4) Amylase

47) Which one of the following antibiotic was extensively used to treat American soldiers wounded in World War-II:-

- (1) Streptokinase
- (2) Penicillin
- (3) Statins
- (4) Neomycin

48) *Propionibacterium sharmanii* is used in the ripening of which cheese ?

- (1) Camembert cheese
- (2) Swiss cheese
- (3) Roquefort cheese
- (4) Limburger cheese

49) How many alcoholic drinks of following are formed by distillation :-

- (A) Wine (B) Beer (C) Whisky
(D) Brandy (E) Rum

- (1) Only one
- (2) Only three
- (3) Only four
- (4) All five types

50) According to the current concept of Central Dogma of molecular biology, information transfer from proteins to nucleic acids does not occur. Which of the following could be considered a violation of the Central Dogma?

- I. Synthesis of DNA on an RNA template by reverse transcriptase.
- II. Gene transcription being activated by a protein binding to regulatory sequences.
- III. Editing of RNA transcripts by the insertion and deletion of ribonucleotides.
- IV. A polynucleotide polymerase specifically synthesizing a gene encoding itself from free deoxyribonucleotides.

- (1) I
- (2) IV
- (3) II and III
- (4) None of the above

51) The bottled juices are clarified by the use of:-

- (1) Pectinases
- (2) Proteases

- (3) Lipases
- (4) Both 1 and 2

52) The coding ratio in all known organisms is three, i.e., three nucleotides specify one amino acid. If DNA were to exclusively consist of only A-T base pairs, what would the minimum coding ratio be assuming that there are only 20 amino acids to be encoded?

- (1) Two
- (2) Three
- (3) Four
- (4) Five

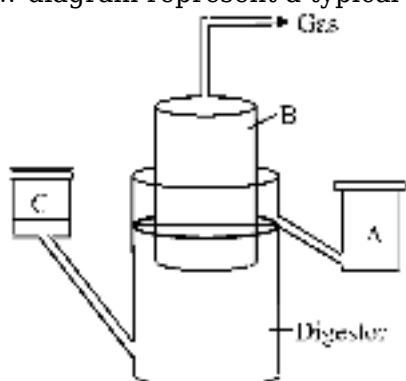
53) Which property is **not** shown by Cyclosporin-A ?

- (1) Immunosuppressive agent
- (2) Clot buster
- (3) Used in organ transplant patients
- (4) Obtain by fungus *Trichoderma polysporum*

54) Lady bird (Beetle with red and black marking) is used as biological control of :

- (1) Mosquito
- (2) Aphids
- (3) Dragonflies
- (4) Both 1 & 2

55) The below diagram represent a typical biogas plant. Select the correct option for A, B and C



respectively.

- (1) A-Sludge, B-Slurry, C- $\text{CH}_4 + \text{CO}_2$
- (2) A-Slurry, B-Sludge, C- $\text{CH}_4 + \text{CO}_2$
- (3) A-Sludge, B- $\text{CH}_4 + \text{CO}_2$, C-Slurry
- (4) A- $\text{CH}_4 + \text{CO}_2$, B-Slurry, C-Sludge

56) Masses of bacteria associated with fungal filaments and form mess like structure known as-

- (1) Flocs
- (2) Guild

- (3) Equilins
- (4) Sludge digester

57) The fungal symbiont in the mycorrhizal associations -

- (a) Absorbs phosphorus from soil
- (b) Provides resistance to stem-borne pathogens
- (c) Provides tolerance to salinity and drought.
- (d) Help to fix free Nitrogen.

The correct statements is / are -

- (1) (a) & (c)
- (2) (a), (c) & (d)
- (3) (a), (b) & (d)
- (4) (a), (b), (c) & (d)

58) The majority of Baculoviruses are used as biological control agents as

- (1) They are species-specific
- (2) They have no negative impact on non-targets
- (3) Beneficial insects are being conserved
- (4) All the above

59) Match the following columns :-

Column-I		Column-II	
A.	Ladybird beetle	(i)	Nitrogen fixation
B.	Mycorrhiza	(ii)	Bio gas
C.	Methanogens	(iii)	Aphids
D.	Rhizobium	(iv)	Glomus

	A	B	C	D
(1)	(ii)	(iv)	(iii)	(i)
(2)	(iii)	(iv)	(ii)	(i)
(3)	(iv)	(i)	(ii)	(iii)
(4)	(iii)	(ii)	(i)	(iv)

- (1) 1
- (2) 2
- (3) 3
- (4) 4

60) Given below are two statements:

Statement I: Microbes like bacteria and many fungi can be grown on nutritive media to form

colonies that are visible to the naked eye.

Statement II: All microbes are harmful and cause diseases in human beings, animals, and plants.
In the light of the above statements, choose the correct answer from the options given below:

- (1) Statement I is false but Statement II is true.
- (2) Both Statement I and Statement II are true.
- (3) Both Statement I and Statement II are false.
- (4) Statement I is true but Statement II is false.

61) Which of the following ways is most likely to decrease the genetic diversity in a population?

- (1) Gene mutation
- (2) Genetic recombination
- (3) Stabilizing natural selection
- (4) Immigration of individuals

62) **Assertion A:** The activated sludge is partly recycled back to the aeration tank to increase the efficiency of the sewage treatment.

Reason R: A significant portion of the activated sludge is used as inoculum to promote the breakdown of organic matter in the sewage.

- (1) Both A and R are true and R is the correct explanation of A
- (2) Both A and R are true but R is not the correct explanation of A
- (3) A is true but R is false
- (4) A is false but R is true

63) **Assertion A:** The probability of allelic sequence variation to be observed in non-coding DNA sequence would be higher.

Reason R: Mutation in non-coding part of DNA may not have any immediate impact in an individual's reproductive ability.

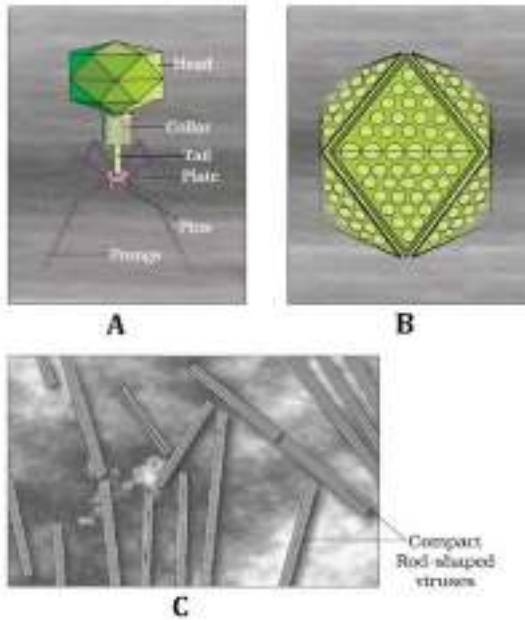
- (1) Both A and R are true and R is the correct explanation of A
- (2) Both A and R are true but R is not the correct explanation of A
- (3) A is true but R is false
- (4) A is false but R is true

64) **Assertion A:** Polymorphism in DNA sequences is the basis of genetic mapping of human genome.

Reason R: Every tissue from an individual show the same degree of DNA polymorphism.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

65) Identify A, B and C in the given diagram.



	A	B	C
(1)	Bacteriophage	TMV	Adenovirus
(2)	TMV	Bacteriophage	Adenovirus
(3)	Bacteriophage	Adenovirus	TMV
(4)	Adenovirus	TMV	Bacteriophage

- (1) 1
- (2) 2
- (3) 3
- (4) 4

66) Relation between amino acid and protein is similar to one that is found in between :-

- (1) Glucose and fructose
- (2) Thymine and uracil
- (3) Nucleosides and nucleic acid
- (4) Nucleotides and nucleic acid

67)

Two individuals A and B have a nucleotide sequence in their DNA-

Individual - A - CCCATACCGAACGAATCGATAC.

Individual - B - CCCATACCGTACGAATCGATAC.

If we find any difference in their DNA, this can be known as-

- (1) SSR
- (2) STR
- (3) SNP
- (4) EST

68) Which of the following absent in DNA-

- (1) Uracil
- (2) Thiamine
- (3) Guanine
- (4) Both 1 & 2

69) Match **List-I** with **List-II**.

List-I		List-II	
(a)	Bacteriophage $\phi \times 174$	(i)	48502 base pairs
(b)	Bacteriophage lambda	(ii)	5386 nucleotides
(c)	<i>Escherichia coli</i>	(iii)	3.3×10^9 base pairs
(d)	Haploid content human DNA	(iv)	4.6×10^6 base pairs

Choose the **correct** answer from the options given below:

- (1) a-i, b-ii, c-iii, d-iv
- (2) a-ii, b-iv, c-i, d-iii
- (3) a-ii, b-i, c-iv, d-iii
- (4) a-i, b-ii, c-iv, d-iii

70) Rho factor terminate-

- (1) Translation by identifying UAA
- (2) Translation by identifying UAG
- (3) Do not identify any t-RNA
- (4) Copying of one type of nucleic acid to another type of nucleic acid

71) In HGP, the last chromosome to be sequenced was the chromosome number _____ in _____.

- (1) 11, May 2003
- (2) 20, Jan 2003
- (3) 1, May 2004
- (4) 1, May 2006

72) A nucleosome is composed of

- (1) Positively charge DNA and negatively charged histone octamer
- (2) A negatively charged H1 histone and positively charged DNA molecule
- (3) Negative charged DNA and positively charged histone
- (4) Negatively charged lysines and positively charged argining amino acids

73) Severo ochoa enzyme is responsible for -

- (1) Template independent enzymatic synthesis of RNA.
- (2) Template dependent enzymatic synthesis of RNA.
- (3) Template independent enzymatic synthesis of DNA.

(4) Template dependent enzymatic synthesis of DNA.

74) Anticodon for 5AGC3

- (1) 5UCG3
- (2) 3UCG5
- (3) 5GCU3
- (4) More than one correct

75) Match the column :-

Column-I		Column-II	
(A)	Termination	(i)	One codon codes for only one amino acid
(B)	Non-ambiguity	(ii)	One amino acid is coded by more than one codon
(C)	Universality	(iii)	Ochre, Amber, Opal
(D)	Degeneracy or redundancy	(iv)	From bacteria to human beings UUU codes for phenylalanine

- (1) A = (iii), B = (ii), C = (iv), D = (i)
- (2) A = (iii), B = (i), C = (iv), D = (ii)
- (3) A = (iv), B = (i), C = (ii), D = (iii)
- (4) A = (iii), B = (iv), C = (i), D = (ii)

76) On translation, mRNA produces a polypeptide chain which contain 50 amino acids. If 26th codon of m-RNA UAU changes to UAG, then :-

- (1) Polypeptide chain will not be changed.
- (2) Two amino acids will be changed.
- (3) Only one amino acid will be changed.
- (4) Polypeptide chain will have only 25 amino acids.

77) **Statement-I:** The organic farmer holds the view that eradication of a creature, even a pest is undesirable.

Statement-II: Predator and parasite can not survive without their prey/host

- (1) Both statement I and statement II are incorrect
- (2) Statement I is correct and statement II is incorrect
- (3) Statement I is incorrect and statement II is correct
- (4) Both statement I and statement II are correct

78) tRNA has the function of

- (1) Transcription
- (2) Adapter for attaching amino acids over mRNA template

- (3) Transferring information to mRNA
- (4) Carry genetic code to cytoplasm

79) Splicing is meant for eliminating -

- (1) Recons
- (2) Mutons
- (3) Exons
- (4) Introns

80)

RNA can acts as

- (a) Genetic material (b) Messenger
- (c) Adapter molecule (d) Structural molecule
- (e) Catalytic molecule

- (1) b,c and e only
- (2) a, d and e only
- (3) a, b and c only
- (4) a, b, c, d and e

81) **A** : Regulation of transcription is the only step for regulation of gene expression in eukaryotes.

B: UTR are found just before the stop codon as well as just after initiation codon

- (1) A & B both are correct
- (2) Only A Correct
- (3) A & B both are incorrect
- (4) Only B correct

82) BGA as a biofertilizer is more advantageous over symbiotic N-fixing bacteria - because-

- (1) BGA fix N into NH_3
- (2) BGA also add organic matter in soil
- (3) Like VAM BGA absorb PO_4
- (4) BGA is unicelled Algae.

83) How many of the following statements are correct w.r.t. genetic code ?

- a. The codon is read in mRNA in a contiguous fashion
- b. All 64 codon are specific for amino acids.
- c. UGG is ambiguous and degenerate codon
- d. Due to degenerate code, prediction of mRNA is not accurate.

- (1) One
- (2) Two
- (3) Three
- (4) Four

84) Find the correct statement w.r.t. transcription:-

- (1) There is single DNA dependent RNA polymerase that catalyses transcription of all types of RNA in plant cell.
- (2) Elongation is controlled by sigma factor.
- (3) Rho factor terminates the transcription, when it associates with DNA polymerase enzyme.
- (4) The promoter and terminator flank the structural gene in a transcription unit.

85) The DNA in nucleoid is organised in :-

- (1) Small loops held by histones
- (2) Large loops held by nucleosomes
- (3) Small loops held by basic proteins
- (4) Large loops held by proteins

86) Which of the following feature is represented by chemically less reactive and structurally more stable nucleic acid ?

- (1) 3' OH group
- (2) Catalytic nature
- (3) Scope of rapid mutation
- (4) Direct involvement in protein synthesis

87) The unequivocal proof that DNA is the genetic material came from the experiment of:-

- (1) Avery, MacLeod and McCarty, they worked with *E.coli*
- (2) Frederick Griffith, he worked with *Streptococcus pneumoniae*
- (3) Avery, MacLeod and McCarty, they worked with Bacteriophage
- (4) Hershey and Chase, they worked with bacteriophage and bacteria

88) Mark correct one w.r.t. euchromatin:

- (1) Transcriptionally inactive
- (2) Densely packed chromatin
- (3) Lightly stained chromatin
- (4) Late replicating

89) BOD is-

- (1) Bio degradable oxygen
- (2) Bio chemical oxygen demand
- (3) Chemical oxygen demand
- (4) Buffet on demand

90) If one strand of DNA has nitrogenous base as 5'-ATCTG-3'. What would be the complementary RNA strand sequence ?

(1) 5'-AUCUG-3'

(2) 5' CAGAU-3'

(3) 3' UAGAC-5'

(4) 3' ATCGU 5'

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	4	4	1	2	1	1	1	1	3	4	3	3	3	2	2	3	1	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	1	3	3	2	2	1	1	4	1	2	1	2	2	1	1	1	4	1
Q.	41	42	43	44	45															
A.	2	3	4	1	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	4	3	1	3	2	1	1	3	3	2	1	3	1	1	4	4	2	2	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	3	1	2	1	3	3	2	3	1	1	3	1	1	2	3	1	3	3	3
Q.	86	87	88	89	90															
A.	4	3	1	1	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	2	2	2	4	3	4	4	4	2	3	2	3	4	4	1	1	2	3	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	2	3	2	4	3	3	2	2	3	2	2	3	3	3	2	3	3	1	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	1	3	3	3	2	2	2	2	2	4	4	2	2	3	1	1	4	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	1	1	2	3	4	3	4	3	4	4	3	1	4	2	4	4	2	4	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	2	2	4	4	1	4	3	2	1										

SOLUTIONS

PHYSICS

1) Minimum rate of burning fuel is $\frac{dm}{dt}$

$$\square mg = u \frac{dm}{dt}$$

$$\text{or, } \frac{dm}{dt} = \frac{mg}{u} = \frac{1000 \times 10}{100} = 100 \text{ kg/s.}$$

2)

$$a = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) g = \frac{g}{8} = g \left(\frac{1 - m_1/m_2}{1 + m_1/m_2} \right)$$

$$1 + \frac{m_1}{m_2} = 8 - \frac{8m_1}{m_2}$$

$$\frac{9m_1}{m_2} = 7 \Rightarrow \frac{m_1}{m_2} = \frac{7}{9}$$

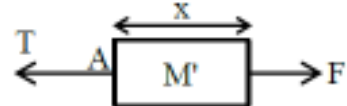
3) $L \rightarrow M$ $a = \frac{F}{M}$

$\frac{M}{L}$

$1 \rightarrow \frac{L}{M}$

$\frac{M}{L}$

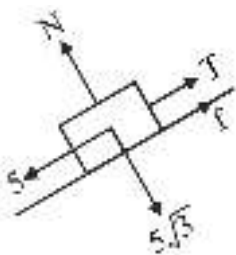
$x \rightarrow \frac{L}{M} \cdot x$



$$F - T = M' a$$

$$F - T = \left(\frac{M}{L} x \right) \left(\frac{F}{M} \right)$$

$$T = \frac{F(L - x)}{L}$$



4)

$$N = 5\sqrt{3}$$

$$f_0 = \mu N$$

$$= 4\sqrt{3} = 6.9$$

$$T = 0$$

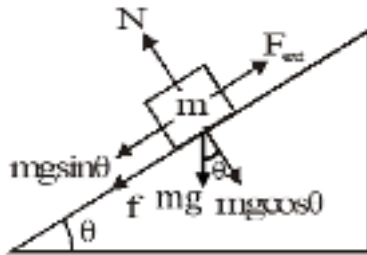
5)

Explain: A block of mass m is on an inclined plane with friction coefficient μ . Find the force required to move the block up the incline.

Concept: Limiting friction.

Formula: Limiting friction force $f_l = \mu N$

Solution:



$$f = \mu N$$

$$f = \mu mg \cos \theta$$

$$F_{\text{ext}} = mg \sin \theta + f$$

$$F_{\text{ext}} = mg \sin \theta + \mu mg \cos \theta$$

Conclusion: Hence option (1) is correct answer.

6) Conceptual

7)

Conceptual

8)

CORRECT ANSWER (1)

$$9) \frac{r_p}{r_d} = \sqrt{\frac{m_p/q_p}{m_d/q_d}}, \frac{r_d}{r_a} = \sqrt{\frac{m_d/q_d}{m_a/q_a}}$$

For proton (p): $m_p = m$, $q_p = e$

For deuteron (d): $m_d = 2m$, $q_d = e$

For a-particle (a): $m_a = 4m$, $q_a = 2e$

$$r_p = \sqrt{\frac{m}{e}}, r_d = \sqrt{\frac{2m}{e}}, r_a = \sqrt{\frac{4m}{2e}} = \sqrt{\frac{2m}{e}}$$

$$r_p : r_d : r_a = 1 : \sqrt{2} : \sqrt{2}$$

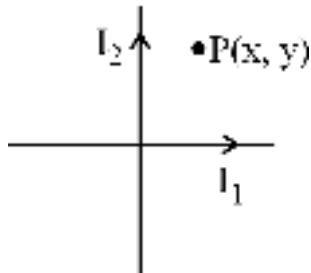
Approximating values, the best matching answer is:

Final Answer: (1) $1 : \sqrt{2} : \sqrt{2}$

10) $B_{\text{arc}} = \frac{\mu_0 I}{8R}$



The direction of the magnetic field is into the plane using right hand thumb rule
Correct option: (1)



11)

If $(B_p)_{\text{net}} = 0$ then

$$|B_1| = |B_2| \text{ at 'P'}$$

$$\frac{\mu_0 I_1}{2\pi y} = \frac{\mu_0 I_2}{2\pi x}$$

$$y = \left(\frac{I_1}{I_2}\right) x$$

12)

$$B = \frac{\mu_0 I}{2\pi r} \left(\frac{r^2 - a^2}{b^2 - a^2} \right)$$

$$\begin{cases} r = \frac{3R}{2} \\ a = R \\ b = 2R \end{cases}$$

13)

Solution:

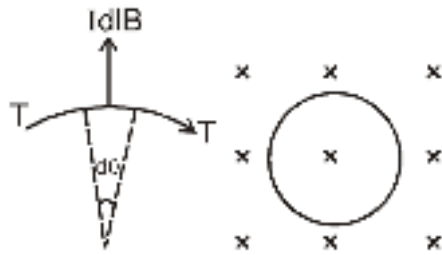
A current-carrying conductor produces only magnetic fields. In the, form of concentric circles.
A moving charge produces both electric and magnetic fields as it carries charge and moves.

Analysis:

Statement-1: True because a current-carrying conductor produces magnetic fields, in the form concentric circles.

Statement-2: False because a moving charge produces both electric and magnetic fields.

14) $\vec{B} = \frac{\mu_0 I}{2\pi R} \odot + \frac{\mu I}{2R} \otimes$



15)

$$2T \sin\left(\frac{d\theta}{2}\right) = IdB = 2T \frac{d\theta}{2} = IRdB$$

$$\Rightarrow T = BIR = \frac{BIL}{2\pi}$$

16)

CORRECT ANSWER (2)

$$17) B_1 = \frac{\mu_0 I_1 \alpha_1}{4\pi R} \quad B_2 = \frac{\mu_0 I_2 \alpha_2}{4\pi R}$$

$$\square I_1 \alpha_1 = I_2 \alpha_2$$

$$\frac{B_1}{B_2} = 1$$

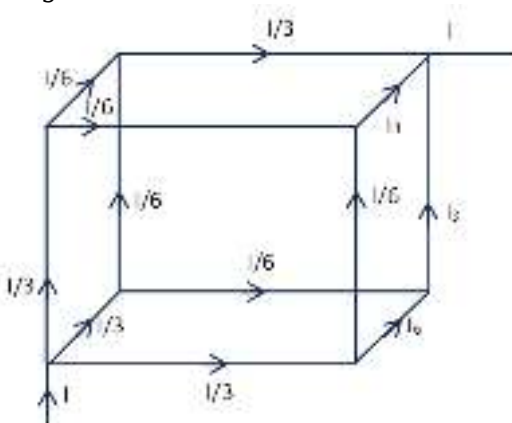
18) **Explaining the question :** In this question a steady current set up cubic network as shown in figure find out the magnetic field field p due to cubic network.

Concept : Magnetic field due to current carrying wire.

$$\text{Formula : } \frac{\mu_0 I}{4\pi r} (\sin\theta_1 + \sin\theta_2)$$

Solution :

Diagram :



The magnetic field due to straight wire of length $\frac{d}{3}$ carrying current $\frac{I}{3}$ at a distance $\frac{d}{2}$ is given by

$$B_{\text{side}} = \frac{\mu_0}{4\pi} \frac{I}{\frac{d}{3}} \left(\frac{d}{2}\right) (\sin 45^\circ + \sin 45^\circ)$$

$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$B_{\text{side}} = \frac{\mu_0 I}{4\pi \frac{d}{2}} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right)$$

$$B_{\text{side}} = \frac{\mu_0 I \sqrt{2}}{6\pi d}$$

The total magnetic field due to opposite sides of the cube each cancel out such that the net magnetic field of point p is zero

Final answer : Option (3)

19) $r_1 : r_2 = 1 : 2$ and $B_1 : B_2 = 1 : 3$

We know that

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi Ni}{r} \Rightarrow \frac{i_1}{i_2} = \frac{B_1 r_1}{B_2 r_2} = \frac{1 \times 1}{3 \times 2} = \frac{1}{6}$$

20) Concept Used:

The magnetic field due to a straight current-carrying wire follows the right-hand rule. Fields are vector quantities, so we sum components using vector addition.

Formulas Used:

- Magnetic field due to a long straight wire:

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2i}{x}$$

- Net magnetic field (vector addition):

$$B_{\text{net}} = \sqrt{(B_x)^2 + (B_y)^2}$$

Calculation in Minimum Steps:

- Given: $B_1 = B_2 = B_3 = B_4 = B$
- Horizontal components: $B_1 + B_3 = 2B$
- Vertical components: $B_2 + B_4 = 2B$
- Net magnetic field:

$$B_{\text{net}} = \sqrt{(2B)^2 + (2B)^2}$$

$$B_{\text{net}} = \sqrt{4B^2 + 4B^2} = \sqrt{8B^2} = 2\sqrt{2}B$$

Final Answer : $B_{\text{net}} = 2\sqrt{2}B$

21) A charged particle moves in a straight line under the action of an electric field whereas it moves in a circular path under the action of a magnetic field. Thus, for the particle moving in a circular path, $\vec{E} = 0, \vec{B} \neq 0$.

22) Here, point P is placed symmetrically with respect to two conductors. The magnetic fields due to the conductors are mutually perpendicular. Their resultant is given by

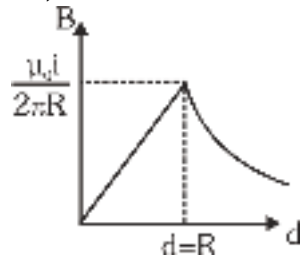
$$B_r = \frac{\mu_0}{2\pi d} \sqrt{(I^2 + I_2^2)}$$

23) $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \sum I$

$$\sum I = 2 - 1 = 1 \text{ A}$$

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0$$

$$24) \quad B = \begin{cases} \frac{\mu_0 i d}{2\pi R^2} : d \leq R \\ \frac{\mu_0 i}{2\pi d} : d > R \end{cases}$$



25)

B is the net B due to all the currents present inside & outside the loop.

$$26) \quad \frac{\mu_0 2I}{2\pi x} = \frac{\mu_0 5I}{2\pi(6-x)}$$

27)

Loop orientation is clockwise

$\therefore \otimes \rightarrow \text{Positive}$ $\odot \rightarrow \text{Negative}$

28) As the solenoids are identical, the currents in Q and R will be same and will be half the current in P. The magnetic field within a solenoid is given by $B = \mu_0 nI$. Hence the field in Q will be equal to the field in 'R' and will be half the field in 'P' i.e. will be 2.0T.

29)

$$B_{\text{end point}} = \frac{\mu_0 nI}{2} = \frac{\mu_0 NI}{2\ell}$$

$$= \frac{4\pi \times 10^{-7} \times 100 \times 2.5}{2(50 \times 10^{-2})}$$

$$= 3.14 \times 10^{-4} \text{ T}$$

$$30) \quad B = \mu^0 \frac{N}{2\pi R_m} \cdot I \quad (\text{given } R_m = 10 \text{ cm})$$

31) By symmetry

$$R_{eq} = 9 \Omega$$

32) As B connected to earth, so potential of B is $V_B = 0$.

Now, current in the given circuit,

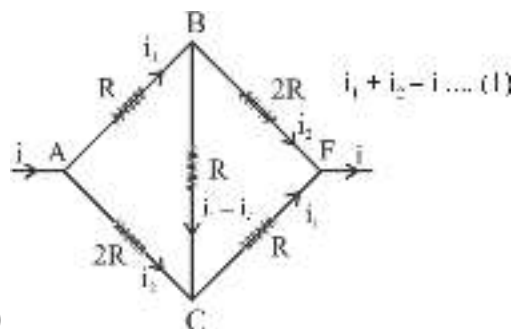
$$i = \frac{50}{5 + 7 + 10 + 3} = 2A$$

Potential difference between A and B is

$$V_A - V_B = 2 \times 12$$

$$\text{or } V_A - 0 = 24$$

$$V_A = 24V$$



33)

loop ABCA

$$-i_1 R - (i_1 - i_2)R + 2i_2 R = 0$$

$$-2i_1 + 3i_2 = 0$$

$$3i_2 = 2i_1 \quad \dots (2)$$

from (1) & (2)

$$i_1 + \frac{2}{3}i_1 = i$$

$$i_1 = \frac{3i}{5} \text{ Put in (1) } i_2 = \frac{2i}{5}$$

$$\text{Current in arm AD} = i_2 = \frac{2i}{5}$$

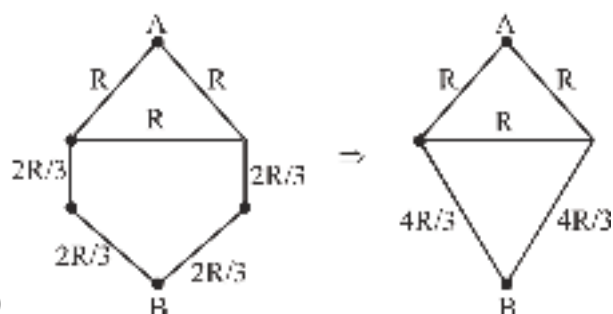
34)

P.d. across cell $V = E - Ir$

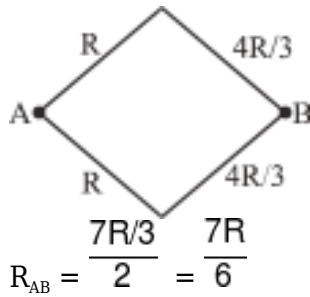
When $I = 0$, $V = E = 2V$

When $V = 0$, $0 = E - 5r \Rightarrow 0 = 2 - 5r$

$$r = 0.4 \Omega$$



35)



36) All bulbs are identical so we can consider them as three identical resistors of value R .
So power consumed $p = i^2 R_{eq}$.

$$(R_{eq})_I = 3R, (R_{eq})_{II} = \frac{2R'}{3}$$

$$(R_{eq})_{III} = \frac{R}{3}, (R_{eq})_{IV} = \frac{3R}{2}$$

37) For maximum current,

$$mR = nr \quad \dots\dots(i)$$

$$\text{Given that } mn = 45 \quad \dots\dots(ii)$$

$$\text{Hence, } m \times 2.5 = n \times 0.5$$

$$\text{or } n = 5m \quad \dots\dots(iii)$$

From equations (ii) and (iii)

$$5m^2 = 45 \text{ or } m = 3$$

$$\square \quad n = 15$$

38)

$$\text{In a row } E_s = (5000) (0.15) = 750 \text{ V}$$

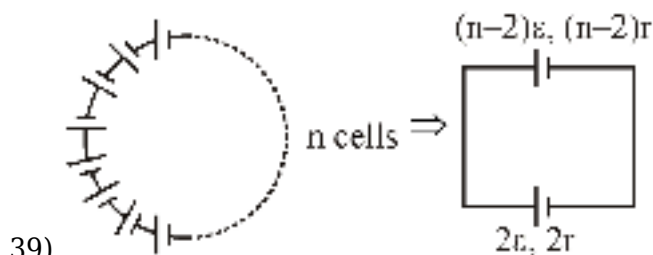
$$r_s = (5000) (0.25) = 1250\Omega$$

for complete circuit

$$E_{net} = 750 \text{ V}$$

$$R_{net} = R + r_{net} = 500 + \frac{1250}{100} = 51205\Omega$$

$$i = \frac{E_{net}}{R_{net}} = \frac{750}{\frac{5}{2.5}} = 1.5A$$



39)

$$I = \frac{(n-2)\varepsilon - 2\varepsilon}{nr} = \left(1 - \frac{4}{n}\right) \frac{\varepsilon}{r}$$

Potential across any one reversed Polarity cell

$$V = \varepsilon + Ir$$

$$= \varepsilon + \left(1 - \frac{4}{n}\right) \varepsilon = 2\varepsilon \left(1 - \frac{2}{n}\right)$$

Hence option (4)

$$40) V_A - (2 \times 2) - 3 - (2 \times 1) = V_B$$

$$\Rightarrow \text{Hence } V_A - V_B = +9 \text{ V}$$

41)

$$I_1 = I_2 + I_3 \quad \dots(1)$$

$$\Rightarrow \frac{50 - V}{5} = \frac{V - 0}{3} + \frac{V - 30}{15} \quad \dots(2)$$

$$150 - 3V = 6V - 30$$

$$\Rightarrow V = 20 \text{ volt}$$

42)

Its a balanced wheat stone bridge

$$\text{then } i_1 = \frac{60}{(60 + 24)} (1.4) = 1 \text{ A}$$

43)

Resistivity is a material property independent of a wire's dimensions.

Since ρ is not dependent on length (or are A , changes in dimensions affect resistance, not resistivity.

ρ = resistivity (depends only on material and temperature)

Final Answer:

The correct option is (4).

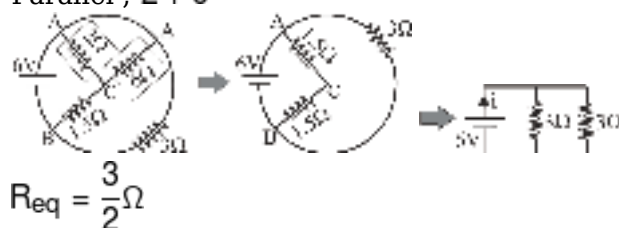
$$44) R = \frac{\rho \ell}{A}, \text{ on stretching volume remains same}$$

$$V = A\ell.$$

$$R = \frac{\rho \ell^2}{V} \Rightarrow \frac{\Delta R}{R} = 2 \frac{\Delta \ell}{\ell}$$

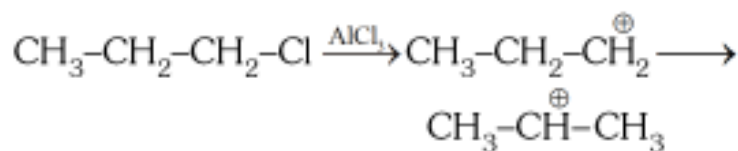
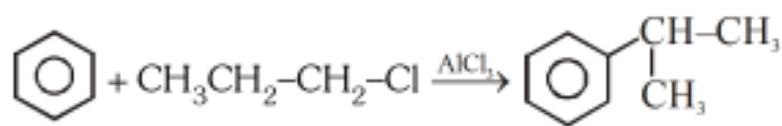
45)

$$\text{Parallel, } \frac{2 \times 6}{2 + 6} = 1.5 \Omega$$



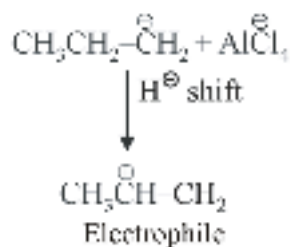
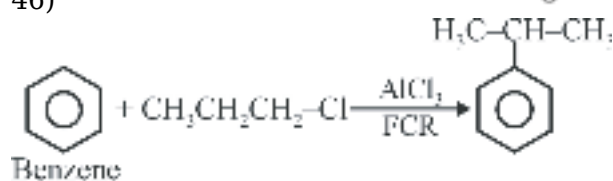
$$i = \frac{6}{3/2} = 4A$$

CHEMISTRY



46)

OR



47)

CORRECT ANSWER (4)

48) NCERT XIth, Pg. # 395

49)

CORRECT ANSWER (1)

50)

FACT

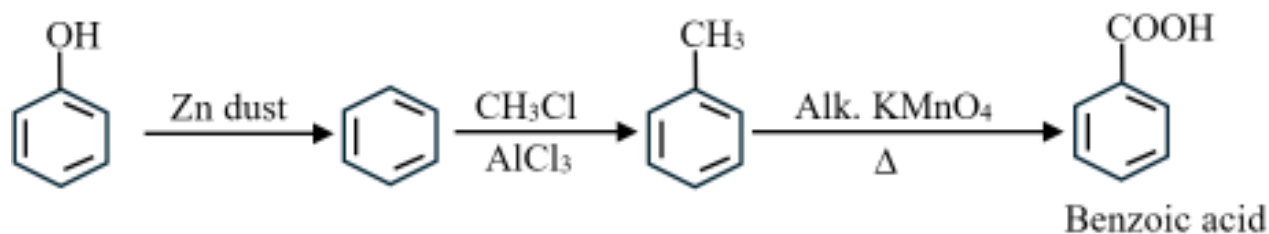
51) **Question Explanation :**

Question is asking about product Z of given sequence

Concept :

Question is based on Reduction of phenol followed by FCR and oxidation.

Solution :



Final Answer :

Option (2)

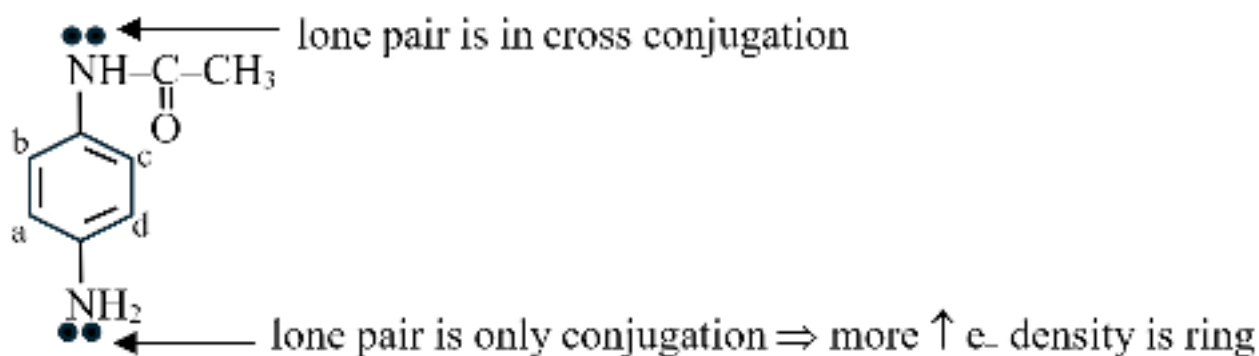
52) Question Explanation :

Question is asking about identify position where ESR is most favourable

Concept :

Question is based -NH_2 donates more as compound by $\text{NH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$.

Solution :



Final Answer :

Option (1)

53) Question Explanation :

Question is asking about how many are ortho and para directing.

Concept :

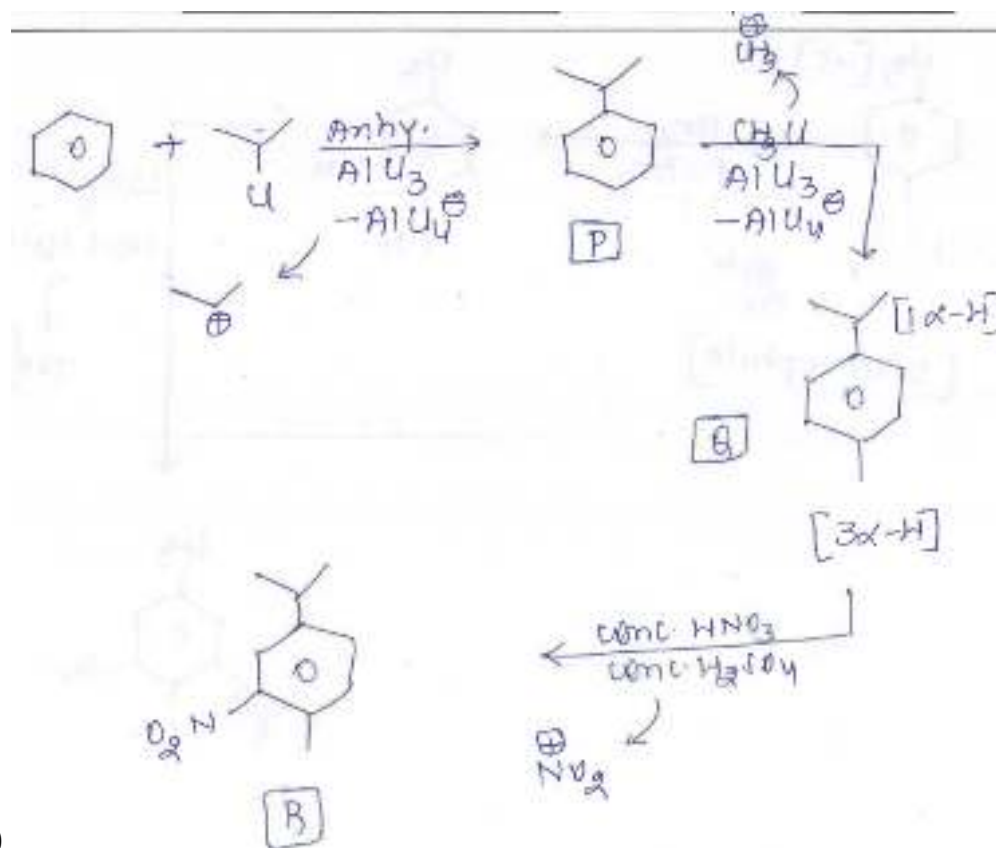
Question is based on -OH , -Cl , are $\text{-NH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$, $\text{-O}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CN}_3$ are ortho and para directing.

Solution :

Lone pair containing groups are ortho and para directing.

Final Answer :

Option (1)



54)

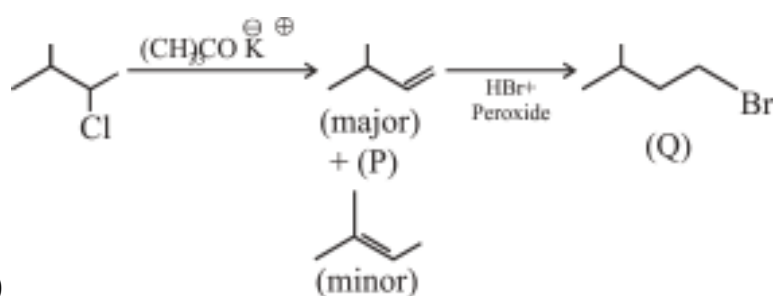
55)

CORRECT ANSWER (3)

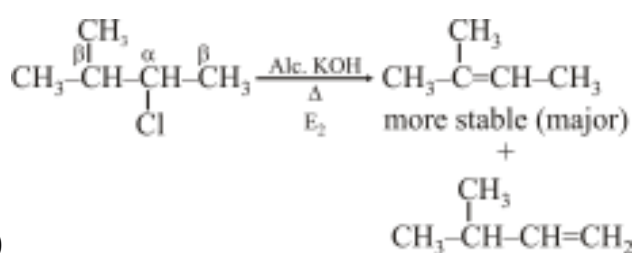
56)

CORRECT ANSWER (2)

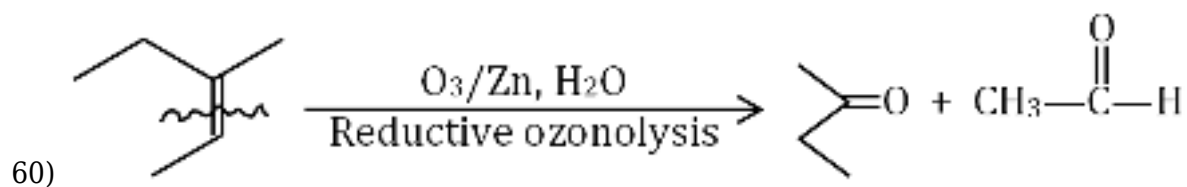
57) Rate of dehydration $\propto$ stability of $\text{C}^\oplus$



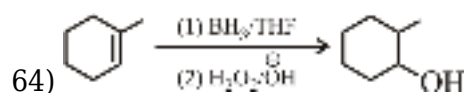
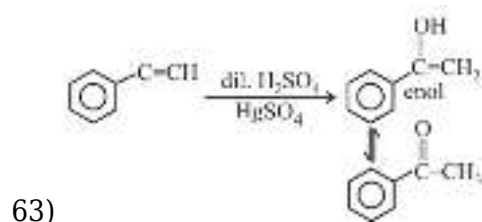
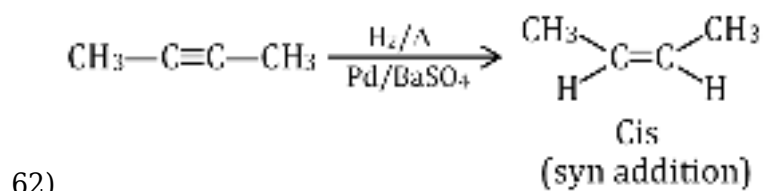
58)



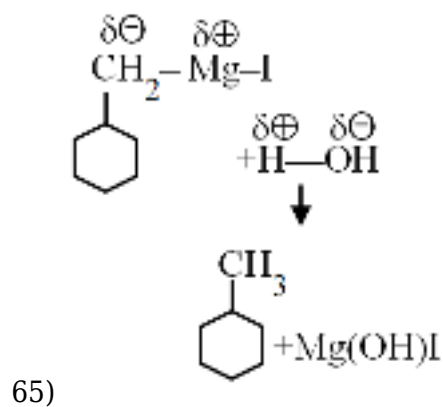
59)



61) Rate of elimination $\propto$ Stability of product



Above reaction is hydroboration oxidation involving anti Markovnikov addition of water.



66)

A = $\text{CH}_3\text{-CH=CH}_2$

B = $\text{CH}_3\text{-CHO}$, C = $\text{CH}_2\text{=O}$

67) Carbocation rearrangement

68)
$$\frac{t_{0.52}}{t_{0.51}} = \left[\frac{A_2^0}{A_1^0} \right]$$

$$\frac{t_{0.5_2}}{1} = \left[\frac{0.5}{2} \right]$$

$$t_{0.5_2} = 0.25h$$

69)

On increasing temperature, molecules with high energy increases i.e., number of activated molecules increases.

70)

Explain the Problem:

The question asks which reaction characteristics change when a catalyst is added to a reaction at a constant temperature.

A. Concept: Effect of catalyst

A. Answer Explanation:

- A. (i) Activation energy: A catalyst lowers the activation energy, thus changing it.
- B. (ii) Equilibrium constant: A catalyst does not affect the equilibrium constant. It speeds up both the forward and reverse reactions equally.
- C. (iii) Reaction entropy: A catalyst does not change the entropy of the reaction. Entropy is a state function that depends on the initial and final states of the reaction, which are not altered by a catalyst.
- D. (iv) Reaction enthalpy: A catalyst does not change the enthalpy of the reaction. Enthalpy is also a state function, and a catalyst does not change the energy difference between reactants and products.

$$71) \text{ROR} = \frac{(\text{ROD})_A}{3} = \frac{(\text{ROD})_B}{2} = (\text{ROA})_C$$

$$72) \frac{r_2}{r_1} = \left(\frac{A_2}{A_1} \right)^x$$

$$\frac{8.4 \times 10^{-6}}{4.2 \times 10^{-6}} = \left(\frac{2}{1} \right)^x$$

$$(2)^x = 2$$

$$x = 1$$

$$\frac{r_3}{r_1} = \left(\frac{B_3}{B_1} \right)^y$$

$$\frac{5.6 \times 10^{-6}}{4.2 \times 10^{-6}} = \left(\frac{0.2}{0.15} \right)^y$$

$$\frac{4}{3} = \left(\frac{4}{3} \right)^y$$

$$y = 1$$

$$r = K[A][B]$$

73)

$t_{1/2}$ is 5 hrs. during the reaction study and it is a first order reaction.

$$74) E_{\text{cell}}^0 = E_{\text{Cu}^{2+}/\text{Cu}}^0 - E_{\text{Cr}^{3+}/\text{Cr}}^0$$

$$E_{\text{cell}}^0 = 0.34 + 0.74$$

$$E_{\text{cell}}^0 = 1.08 \text{ V}$$

$$75) 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$$

$$\therefore E = E^0 - \frac{0.0591}{2} \log \frac{P_{\text{H}_2}}{[\text{H}^+]^2}$$

$$0 = 0 - 0.0295 \log \frac{P_{\text{H}_2}}{(10^{-7})^2}$$

$$\frac{P_{\text{H}_2}}{(10^{-7})^2} = 1$$

$$P_{\text{H}_2} = 10^{-14} \text{ atm}$$

76)

Question Explanation:

The question requires an evaluation of an assertion regarding the effect of copper ion concentration on the EMF of a Daniel cell and a provided reason based on the Nernst equation.

Given Data

Assertion: Increasing the concentration of Cu^{2+} ions in the copper half-cell of a Daniel cell increases the cell's EMF.

Reason: The Nernst equation for the EMF of a Daniel cell is

$$E_{\text{cell}}^0 = E_{\text{cell}}^0 + \frac{0.059}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$$

Concept: Nernst Equation.

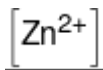
Solution Explanation:

The Nernst equation for a Daniel cell ($\text{Zn}_{(\text{s})} + \text{Cu}^{2+}_{(\text{aq})} \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu}(\text{s})$) is

correctly given by the Reason as

$$E_{\text{cell}}^0 = E_{\text{cell}}^0 + \frac{0.059}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$$

If the concentration of Cu^{2+} ions, i.e., $[\text{Cu}^{2+}]$, increases, the ratio $\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$ increases.



0

Consequently, the term $\log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$ increases. Since this term is added to E_{cell} (which is constant for a given cell at a specific temperature), the overall E_{cell} increases. Thus, the Assertion is True, and the Reason correctly explains the Assertion through the Nernst equation.

Final Answer: (1)

77)

CORRECT ANSWER (3)

78)

CORRECT ANSWER (1)

79)

CORRECT ANSWER (1)

80)

CORRECT ANSWER (2)

81)

CORRECT ANSWER (3)

82)

1. Calculate the Conductivity (κ)

Conductivity (κ) is related to resistance (R) and the cell constant (G^*) by the following equation:

$$\kappa = G^* / R$$

Given:

A. $R = 250 \, \Omega$

B. $G^* = 1.15 \, \text{cm}^{-1}$

$$\kappa = 1.15 \, \text{cm}^{-1} / 250 \, \Omega = 0.0046 \, \Omega^{-1} \, \text{cm}^{-1} \text{ or } 0.0046 \, \text{S cm}^{-1} \text{ (Siemens per centimeter)}$$

2. Calculate the Equivalent Conductance (Λ_{eq})

Equivalent conductance (Λ_{eq}) is related to conductivity (κ) and normality (N) by the following equation:

$$\Lambda_{eq} = (\kappa \times 1000) / N$$

Given:

A. $\kappa = 0.0046 \text{ S cm}^{-1}$

B. $N = 1 \text{ N}$

$$\Lambda_{eq} = (0.0046 \text{ S cm}^{-1} \times 1000) / 1 \text{ N} = 4.6 \text{ S cm}^2 \text{ eq}^{-1} \text{ or } 4.6 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$$

Therefore, the equivalent conductance of the CH_3COOH solution is $4.6 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$, which corresponds to option (1).

$$\begin{aligned} \Lambda_{eq} &= \frac{K \times 1000}{N} \\ \neq K &= G \times G \\ &= \frac{1}{250} \times 1.15 = 0.0046 \pi^{-1} \text{ cm}^{-1} \\ \Lambda_{eq} &= \frac{0.0046 \times 1000}{1} = 4.6 \pi^{-1} \text{ cm}^2 \text{ eq}^{-1} \end{aligned}$$

83)

$$\lambda_{eq} = \frac{1000 \times \kappa}{N}$$

84)

CORRECT ANSWER (3)

85)

Question Explanation:

The question asks us to calculate the molar conductance at infinite dilution for acetic acid (CH_3COOH).

Given Data:

A. $\lambda_m^\infty(\text{HCl}) = 425 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$

B. $\lambda_m^\infty(\text{NaCl}) = 188 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$

C. $\lambda_m^\infty(\text{CH}_3\text{COONa}) = 96 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$

Concept: Kohlrausch's Law

Mathematical Calculation:

We can manipulate the given equations to find $\lambda_m^\infty(\text{CH}_3\text{COOH})$:

$$\lambda_m^\infty(\text{CH}_3\text{COOH}) = \lambda_m^\infty(\text{CH}_3\text{COONa}) + \lambda_m^\infty(\text{HCl}) - \lambda_m^\infty(\text{NaCl})$$

$$\lambda_m^\infty(\text{CH}_3\text{COOH}) = (96 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}) + (425 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}) - (188 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1})$$

$$\lambda_m^\infty(\text{CH}_3\text{COOH}) = 521 - 188 = 333 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

Final Answer:

The molar conductance at infinite dilution for acetic acid is $333 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$.

Correct option (3).

86)

CORRECT ANSWER (4)

87)

Question Explanation:

The question asks for the reason why a saturated solution of KNO_3 is used in making a salt bridge for electrochemical cells.

Given Data Statement: Saturated solution of KNO_3 is used to make a 'Salt-bridge'.

Concept : Function of a Salt Bridge and Ionic Mobility

Solution/Explanation: A salt bridge completes the electrical circuit by allowing ion migration and maintains electrical neutrality in the half-cells. For this to occur effectively and to minimize the liquid junction potential, the electrolyte in the salt bridge should have cations and anions with similar ionic mobilities or velocities. Potassium nitrate (KNO_3) is chosen because the velocities of K^+ ions and NO_3^- ions are nearly the same. High solubility (Option 4) is a prerequisite for making a concentrated solution but the key functional reason for its choice over other soluble salts lies in the comparable ionic velocities.

Final Answer: (3)

88)

Question Explanation;

The problem requires determining the standard electrode potential for the $\text{Fe}^{3+}/\text{Fe}^{2+}$ couple ($E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}}$) given the standard electrode potentials for Fe^{2+}/Fe (x_1) and Fe^{3+}/Fe (x_2).

Given Data

A. $E^\circ_{\text{Fe}^{2+}/\text{Fe}} = x_1$ Reaction: $\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Fe}(\text{s})$

B. $E^\circ_{\text{Fe}^{3+}/\text{Fe}} = x_2$ Reaction: $\text{Fe}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Fe}(\text{s})$

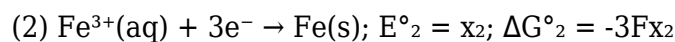
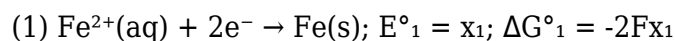
Concept :

Relationship between Standard Gibbs Free Energy Change (ΔG°) and Standard Electrode Potential (E°): $\Delta G^\circ = -nFE^\circ$, where n is the number of electrons transferred and F is Faraday's constant. ΔG° is an extensive property and can be added or subtracted.

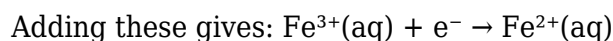
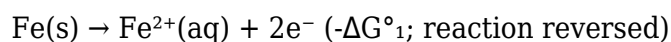
Mathematical Calculation/Reasoning:

Let the required reaction be: (3) $\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{aq})$; $E^\circ_3 = ?$; $\Delta G^\circ_3 = -1FE^\circ_3$

The given reactions are:



To obtain reaction (3), we can write it as Reaction (2) - Reaction (1): $\text{Fe}^{3+}(\text{aq}) + 3\text{e}^{-} \rightarrow \text{Fe}(\text{s})$ (ΔG°_2)



$$\text{So, } \Delta G^{\circ}_3 = \Delta G^{\circ}_2 - \Delta G^{\circ}_1$$

$$-1F E^{\circ}_3 = (-3F x_2) - (-2F x_1)$$

$$-F E^{\circ}_3 = -3F x_2 + 2F x_1$$

$$\text{Dividing by } -F: E^{\circ}_3 = 3x_2 - 2x_1$$

Final Answer : (1)

89)

Question Explanation :

The question requires calculating the pH of a buffer solution using the given reduction potential of a hydrogen electrode immersed in it.

Given Data Reduction potential of hydrogen electrode (E) = -0.591 V.

Concept :

Nernst Equation for Hydrogen Electrode

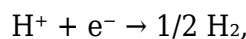
Mathematical Calculation/Reasoning :

The Nernst equation for the hydrogen electrode

$(2\text{H}^{+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{H}_2(\text{g}))$ at 25°C is:

$$E = E^{\circ} - (0.0591/n) \log ([\text{H}_2]/[\text{H}^{+}]^2)$$

Assuming standard conditions for H_2 (pressure = 1 atm) and $E^{\circ} = 0 \text{ V}$ for the standard hydrogen electrode (SHE), and considering the reaction as



$$n=1: E = E^{\circ} - (0.0591/1) \log (1/[\text{H}^{+}])$$

$$E = 0 - 0.0591 (-\log[\text{H}^{+}])$$

$$E = -0.0591 \text{ pH}$$

$$\text{Given } E = -0.591 \text{ V: } -0.591 \text{ V} = -0.0591 \text{ pH}$$

$$\text{pH} = -0.591 \text{ V} / -0.0591 \text{ V.pH}$$

pH = 10.

Final Answer : (1)

90)

Question Asking About:

Number of Electrons transfer, E°_{cell} are given and Equilibrium constant is to be found.

Given Data:

$$E^\circ_{\text{cell}} = 0.591$$

No. of electrons (n) = 2

Concept :

The relationship between standard EMF.

Formula :

$$- E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n} 10 \log \frac{[P]}{[R]}$$

At equilibrium: $E_{\text{cell}} = 0$, $\frac{[P]}{[R]} = K_{\text{eq}}$

Solution/Explanation/Calculation:

$$0.591 = \frac{0.0591}{n} 10 \log K_{\text{eq}}$$

$$10 \log K_{\text{eq}} = 20$$

$$K_{\text{eq}} = 10^{20}$$

Hence,

Option (2) is correct

Level : Tough

BIOLOGY

91)

CORRECT ANSWER (4)

92)

CORRECT ANSWER (2)

93) Explanation :

LSD (Lysergic acid diethylamide) is a powerful hallucinogenic drug that is obtained from *Claviceps purpurea*.

Answer Option is (2).

94)

CORRECT ANSWER (2)

95)

CORRECT ANSWER (4)

96) NCERT XII, Pg. No. 159 Para-8.5

97)

CORRECT ANSWER (4)

98)

CORRECT ANSWER (4)

99)

CORRECT ANSWER (4)

100)

CORRECT ANSWER (2)

101)

CORRECT ANSWER (3)

102)

CORRECT ANSWER (2)

103)

NCERT-XII, Pg. # 150

104) NCERT, Pg # 154

105)

Colostrum is nutritious contains proteins, several antibody, essential fatty acids for proper development of baby.

106) NCERT XII Pg # 60, 61

107) NCERT-XII, Pg. # 63-64

108)

CORRECT ANSWER (2)

109) NCERT Pg. # 64

110)

CORRECT ANSWER (3)

111)

CORRECT ANSWER (1)

112) Ncert Pg No: 44

113) Ncert Pg No: 43

114) Ncert Pg No: 42 - 45

115) Ncert Pg No: 44

116)

CORRECT ANSWER (3)

117)

CORRECT ANSWER (3)

118)

CORRECT ANSWER (2)

119)

CORRECT ANSWER (2)

120)

CORRECT ANSWER (3)

121)

CORRECT ANSWER (2)

122)

CORRECT ANSWER (2)

123)

CORRECT ANSWER (3)

124)

CORRECT ANSWER (3)

125)

CORRECT ANSWER (3)

126)

CORRECT ANSWER (2)

127)

CORRECT ANSWER (3)

128) NCERT-XII Pg. # 43/46(H) Fig:3.1

129)

The correct answer is **Option 1: 250.**

Each testis in the human male reproductive system is divided into approximately **250 testicular lobules**. These lobules are cone-shaped compartments that contain the seminiferous tubules, where sperm production occurs.

130)

CORRECT ANSWER (4)

131)

CORRECT ANSWER (3)

132)

CORRECT ANSWER (1)

133) NCERT XII, Page # 35

134)

CORRECT ANSWER (3)

135)

CORRECT ANSWER (3)

136) **Explanation:**

Monascus purpureus is a fungus that produces **statins**, which are compounds used to lower blood cholesterol levels.

The correct answer is **2. Statins.**

137) NCERT (XII) Pg. # 182

138) NCERT (XII) Pg. # 181

139) NCERT (XII) Pg. # 152

140) NCERT (XII) Pg. # 183

141)

NCERT-XII, Pg. # 183

142)

NCERT-XII, Pg. # 182

143)

Clot buster

144) The correct answer is 2: **Aphids. Explanation :** The ladybird (beetle with red and black markings) is used as a biological control for **aphids**. Ladybirds are natural predators of aphids, helping control their population in crops and gardens.

145) NCERT (XII) Pg. # 186

146) NCERT (XII) Pg. # 186

147) NCERT XII Pg. # 158

148) The correct answer is: **4. All the above**

Explanation: Baculoviruses are used as biological control agents because they are **species-specific**, meaning they target only specific pests without affecting other organisms. They also have **no negative impact on non-target species**, and their use helps in **conserving beneficial insects**. Therefore, all the given statements are true.

149)

NCERT-XII, Pg. # 156,157,158

150) Statement I: Correct. Bacteria and many fungi can indeed be grown on nutritive media to form colonies that are visible without the aid of a microscope, as stated in the text. Statement II: Incorrect. While microbes can cause diseases, they are not all harmful; many microbes are beneficial to humans in various ways, including in medical and industrial applications, which contradicts the assertion that all microbes are harmful.

151)

CORRECT ANSWER (4)

152) The assertion is true as a small part of the activated sludge is indeed recycled back to the aeration tank to maintain the population of the microorganisms that break down the organic matter, thereby increasing the efficiency of the sewage treatment process. The reason is also true and it is the correct explanation for the assertion because the recycled sludge acts as an inoculum, which contains microorganisms that are necessary for the degradation of organic matter in the sewage.

153)

CORRECT ANSWER (1)

154)

CORRECT ANSWER (2)

155)

The correct answer is Option 3:

- A. **A - Bacteriophage:** The image shows a bacteriophage with its characteristic head, tail, and tail fibers.
- B. **B - Adenovirus:** The image depicts a non-enveloped virus with a symmetrical, icosahedral capsid, which is characteristic of adenoviruses.
- C. **C - TMV (Tobacco Mosaic Virus):** The image displays rod-shaped viruses, which is the typical morphology of TMV.

156) NCERT XI Pg. # 149

157) NCERT-XII, Page # 101

158)

CORRECT ANSWER (4)

159) NCERT-XII, Page # 80

160) NCERT XII, Page # 168

161) **Solution:**

The correct answer is Option 4: 1, May 2006.

Explanation: In the Human Genome Project (HGP), the last chromosome to be sequenced was chromosome 1, and this was completed in May 2006. Chromosome 1 is the largest human chromosome, and sequencing it was a significant milestone in the HGP, marking the completion of the full human genome sequence.

So, the correct answer is 1, May 2006.

162) **Explanation:**

- A. A nucleosome is the basic unit of DNA packaging in eukaryotic chromatin.
- B. It consists of negatively charged DNA, which wraps around a positively charged histone octamer.
- C. Histones are rich in basic amino acids like lysine and arginine, giving them a positive charge that helps them bind tightly to negatively charged DNA.

Since DNA is negatively charged and histones are positively charged, **the correct answer is Option 3.**

163) NCERT-XII, Pg. No. # 112 [E]

NCERT-XII, Pg. No. # 120 [H]

164)

NCERT-XII Pg# 112(E) & 120(H)

165) NCERT-XII Pg. # 112

166)

Polypeptide chain will have only 25 amino acids.

167) NCERT-XII Pg. # 117

168)

NCERT XII Pg.# 114

169) NCERT-XII, Pg. # 95

170) **Solution:**

The correct answer is Option 4: a, b, c, d and e.

Explanation: RNA can perform multiple functions, and it plays a vital role in various biological processes:

- A. (a) Genetic material: RNA can act as genetic material in some viruses (e.g., retroviruses like HIV), where RNA carries genetic information instead of DNA.
- B. (b) Messenger: mRNA (messenger RNA) is the template for protein synthesis and carries genetic information from DNA to ribosomes.
- C. (c) Adapter molecule: tRNA (transfer RNA) acts as an adapter molecule that helps in translating the genetic code into proteins by bringing the appropriate amino acids to the ribosome.
- D. (d) Structural molecule: rRNA (ribosomal RNA) is a structural component of ribosomes, which are essential for protein synthesis.
- E. (e) Catalytic molecule: Some RNA molecules, known as ribozymes, have catalytic properties and can catalyze specific biochemical reactions, similar to enzymes.

Thus, RNA can act as all of the mentioned options, making Option 4 the correct answer.

171) NCERT Pg # 116

172) NCERT-XII, Pg. # 101

173) NCERT Pg # 112

174) NCERT Pg # 110

175)

NCERT Pg # 94

176) NCERT Pg # 103

177) NCERT-XII, Pg. # 86

178) NCERT Pg # 100

179) NCERT XII, Pg. # 116

180) NCERT Pg. # 108 (1st Para)